Recovering Observer Spacetime and Einstein Dynamics from Overlap Consistency

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Abstract

This paper develops two layers of Observer Patch Holography: observer clocks, causal order and spatial geometry read from the same repair process, and Einstein dynamics as the coarse-scale response of that process. Normalized repair response on the twelve-port carrier gives a rank-three metric quotient and, with an adjoined scalar line, a 1+3 Lorentz carrier. Conservative Fibonacci record populations with complete reads inside a shrinking radius and a declared model clock have causal orders and normalized counts converging to flat $1 + 3$ spacetime; the ordering fraction of interior causal diamonds tends to $1/10$, and interval counts reconstruct proper-time ratios. Boost invariance selects the light cone of the declared quadratic form among closed convex pointed cones, and finite null tomography reconstructs a symmetric tensor from null projections. Null balance on nine design directions, with Ward and Bianchi identities, gives the Einstein equation up to a cosmological constant, and the small-ball entanglement-equilibrium calculation supplies the required tensor response under six realization premises. The proved centered Weyl vacuum supplies a conserved Lorentz stress tensor, its geometric double-cone modular generator and a finite vacuum-relative first law. For the exact free lattice source, compact state perturbations and finite quadratic stress readouts converge to the same leading stress-response matrix with explicit error bounds, closing a specified free-sector matter-response premise. Its join with an area law, entropy stationarity and responsive geometry on one refinement tower remains to be constructed; the static radial geometry of the executable construction fails null balance for its own scalar excitation.

Interpretation and state variables. OPH proposes one structure with two descriptions: a timeless, self-referential complete history and its actual internal computation. Observers are proposed to reverse-engineer their reality's substrate and construct hardware whose execution realizes that same history, including its builders and hardware, in emergent, mutually calibrated local times. The candidate must supply construction, initialization, controls, memory, read/write and record services, clock comparisons, and their resource costs; no external host or additional time is assumed. Its realization map must specify whether it matches one actual history, a family of preparation and intervention responses, or every independently allowed microstate. These are different requirements. A candidate must specify the history domain and closure operator; neither the schematic equation nor the finite constructions establish this physical realization or select a unique universe. An actual quantum record history also need not assign simultaneous outcomes to incompatible unperformed measurements.

The proposed primitives are patch states, records, interfaces, and admissible operations. Physical space, duration, and matter are intended emergent readouts. A supplied carrier incidence, geometric domain, action, representation, or clock is a premise of the construction using it. Complete-history closure, an instantaneous repair fixed point, a public normal form, and a stationary probability law are different objects. A stationary law can support changing individual histories. An operational observer is defined by its access and record interfaces; consciousness requires an additional psychophysical interpretation.

Basis, premise labels and paper set. OPH has one hypothesis, closure: the universe is a timeless self-referential structure whose observers recover its specification from their records and realize it. Its internal-realization policy requires its operations and information access to be accounted within the structure; operational agreement on accepted shared data defines what is public. The executable specification of carrier, repair law, observer algebra and closure inputs is implemented and runs; it is a candidate judged by whether it reproduces what observers measure. The labels A1, A2 and A3 name theorem premises: the carrier, federation and response layer of that specification; the agreement condition with an additional response-endogeneity principle; and an adopted least-informative state-selection principle. Closure alone implies neither the entropy minimizer nor inner implementation of rechartings on the connected response algebra. A flagship synthesis gives the account, an executable observer construction states the specification with its acceptance criteria and premise map, and the theorem library carries conditional implications whose antecedents that premise map tracks.

Time conventions. Host runtime, solver counters, native attempt counts, local event-clock readings, event precedence or logical numbering, model coordinates or lapse, modular parameters, and calibrated observer-clock readings have distinct roles. Their identification requires explicit maps. A modular parameter is dimensionless; its geometric and operational readings require the stated branch and clock premises. Consistent local clock comparisons require no preferred universal clock or identical clock rates. Shared time parameters in restricted benchmarks are declared model choices. Experienced succession and a final instantaneous cosmic endpoint do not follow from a modular-flow or finite-convergence theorem.

This paper's hypothesis and place in the tower

This paper sits at layers 4 and 6 of the tower. Its hypothesis is that the clocks, causal order and spatial geometry an observer reads are produced by the same native repair process that runs the executable observer [1], and that Einstein dynamics is the coarse-scale response of that process to its own matter.

The paper proves finite and analytic theorems under displayed premises. The rank-three source Gram quotient and its $1+3$ Lorentz carrier are exact. On supplied conservative Fibonacci populations with a complete local read law and model clock, the causal order and counts converge to flat $1+3$ spacetime and a count clock reads proper-time ratios. Cap modular covariance gives Lorentz kinematics on the support-visible geometric subnet. The gravitational theorem is an implication. Null balance on nine design directions, together with Ward and Bianchi identities, gives $G_{ab}+\Lambda g_{ab}=8\pi G\langle T_{ab}\rangle$ (Figure 1); the physical response step follows from the following realization premises on one refinement tower:

(G1) a Lorentzian event and geometry domain with the required locality, differentiability, causal and count/volume limits;

(G2) geometrically normalized cap modular flow whose directional charges define the same matter stress tensor used in propagation and interactions;

(G3) its Ward identity, independent of the Einstein equation;

(G4) the edge-center entropy split and area normalization, the modular first law, and generalized-entropy stationarity at fixed proper volume;

(G5) controlled small-ball remainders and sufficient observer and null directions;

(G6) universal coupling, a common vacuum reference and one positive Newton constant.

Realizing (G1)–(G6) on one refinement tower of the executable model is the open join of this paper. The following facts bound it. The static radial product, the one geometry tested on the executable construction, has vanishing radial null Einstein contractions while an admitted scalar excitation has positive relative null stress, so it cannot balance both; a responsive geometry is required. Stationary Markov or record entropy is not the fixed-volume generalized-entropy variation that (G4) requires (Section B.5). The finite regional calculations supply a bulk/edge entropy split and a quantitative noncommuting first law on native return histories, two-sided entropy variations of one prepared interacting region, and a complete native action of that region's modular polynomial (Sections B.6 to B.8); their modular parameter is dimensionless and carries no area or stress identification. The native screen generator A has spectral dimension two, and the $1+3$ scalar description uses a declared radial extension. The Fibonacci population, read law and model clock of the causal limit are premises until the native process produces them.

A proof that no responsive geometry on the native process balances the null stress of its own admitted excitations, or that its cap modular flow or record entropy cannot meet (G2) or (G4) on any refinement tower, would refute this gravitational branch. A refinement tower on which every component premise holds while the composed Einstein relation fails would contradict the stated implication.

Derivation map

Normalized repair response gives one positive rank-three source Gram quotient E. Signed cumulative records and conservative seam words embed densely in its Euclidean completion. Its unit directions form S^2; adjoining a scalar line gives $\mathbb{R}\oplus E$ with quadratic form $t^2-\|x\|^2$. This connects the source metric to a four-dimensional Lorentz carrier without selecting a physical event clock. The conditional implications on this carrier are

$$\begin{aligned}
\text{nonzero closed convex pointed cone}\\ \text{finite irreducible operational rotations}\\ \text{one continuous boost direction}
&\Longrightarrow \text{Lorentz displacement cone}\\ &\quad\text{up to time orientation},\\
\text{conservative Fibonacci populations}\\ \text{complete local reads and } \Delta{=}a/c
&\Longrightarrow \text{flat causal-order limit}\\ &\quad\text{raw count–volume convergence},\\
\text{controlled order/count limit}\\ \text{accessible interval records}\\ \text{one reference interval}
&\Longrightarrow \text{proper-time ratios from counts},\\
\text{positive translations and finite null tomography}
&\Longrightarrow \text{conditional conserved } \langle T_{ab}\rangle,\\
\text{physical stress and universal coupling}\\ \text{entropy, small-ball and tensor-curvature limits}
&\Longrightarrow G_{ab}+\Lambda g_{ab}=8\pi G\langle T_{ab}\rangle.
\end{aligned}$$

These implications have distinct hypotheses. Operational boost covariance acts on admissible histories and interventions; an algebraic action on the celestial sphere does not establish it. Finite proper carrier rotations and one continuous boost of the declared quadratic form select its light

cone with unit speed in the scalar coordinate (Theorem A.9). The signature and the time axis are declared, so the theorem selects the invariant cone of a chosen Lorentz form rather than deriving Lorentz structure.

Causal order and counts. Put $\varphi = (1+\sqrt{5})/2$ and $L = 2/\sqrt{\varphi+2}$. For consecutive Fibonacci numbers $q = F_n$ and $p = F_{n+1}$, $n \geq 4$, the q^3 spatial records have a source-metric assignment to equal-volume cells with error $H_q = 2\sqrt{3}L/q$. Each event reads every preceding-layer record inside radius $a_q = L/\sqrt{q}$, including its own preceding record, and the model layer duration is $\Delta_q = a_q/c$, with $c > 0$. Segment subdivision bounds the reachable radius between $c(1-2h_q/a_q)T$ and cT; since $h_q \leq H_q$ and $H_q/a_q \to 0$, these cones agree in the limit away from their null boundary. Normalized raw counts converge to $dt\, d^3x$ volume (Proposition A.4), and on fixed interior timelike intervals the ordering fraction tends to 1/10. The finite path construction, the layered order, the count-clock enclosure, the generated-interval and strict-pair count limits and the 1/10 limit for diamonds inside the sampling window are formalized in Lean; the general moving-tip volume-error estimate is an analytic proof. The family is a discretization of flat $1+3$ space, with the ball read as a light cone and a layer as a time slice, so its limit theorems confirm the discretization rather than select the dimension: the spatial dimension three comes from the rank-three carrier, and the time axis is adjoined. Under the same count limit, the observer readout $(|I|/|J|)^{1/4}$ converges to the proper-time ratio of two interior timelike diamonds; it reads accessible ancestry and counts, with a reference interval fixing units (Theorem A.10, Proposition A.11). Finite readings of the family are in Appendix A.

Exact order embeddings form a separate finite interface. For a source spatial readback and positive height scale, a parent-edge speed bound maps authenticated ancestry into the future cone; equal-height spatial separation and strict spacelike separation of increasing-height nonancestors give the converse, hence an injective order embedding. The source height is the attained longest authenticated-parent-chain length. Height is not identified with proper time, or element number with volume; a causal-set faithful embedding also requires the appropriate population density[4, 5]. Under the continuum causality hypotheses, causal order and volume determine conformal geometry and scale[6, 7].

Source metric. For the declared twelve-port repair mean, the centered equal-port response kernel has an exact spectral decomposition. Dividing by its slow response before the infinite-step limit selects a rank-three projector whose range, with the induced Gram form, is an intrinsic Euclidean carrier. Every finite-step kernel has rank eleven on the centered space and rank six on the signed sector. On the signed cumulative port-record/load module the limiting form has a three-dimensional radical on the real six-control extension; the integer module has no nonzero metric kernel, embeds densely in the quotient, and completes to the same carrier up to the unique label-preserving isometry. The source load map is onto $\mathbb{Z}^6$, with the antipodally equal loads as its kernel. Conservative seam boundaries have image

$$D_6 = \{z \in \mathbb{Z}^6 : \textstyle\sum_i z_i \equiv 0 \pmod 2\},$$

with Smith invariants $(1,1,1,1,1,2)$; the pullback response metric makes this submodule dense in the same completion. Cumulative record addition acts simply transitively on the record carrier and by isometries on its completion. The sixty directed seam labels form one proper carrier-rotation orbit. The thirteen finite covariance and isometry table identities of the proper-action subpacket are checked with Lean's `native_decide`; they trust the native compiler and runtime as well as the kernel and are not kernel-only receipts. The source metric does not choose a preferred Cartesian frame or identify record addition with physical translation.

Finite tests for chart transitions. Let u_p be the twelve exact source vectors, of squared radius $R^2 = \varphi + 2$, and declare $k_p = (1, u_p)$ and $G = \mathrm{diag}(-R^2, 1, 1, 1)$. The source tensor decoder has rank nine and kernel $\mathbb{R}G$ on symmetric four-matrices. Thus a supplied invertible real map L passes all twelve null tests $(Lk_p)^T G(Lk_p) = 0$ exactly when

$$L^T G L = \lambda G \qquad \text{for a unique } \lambda > 0.$$

The kernel gives proportionality; inertia gives positivity. Normalizing by $\sqrt{\lambda}$ gives a Lorentz isometry. Future preservation requires $L_{00} > 0$, and properness requires $\det L > 0$ separately: parity and time reversal pass the null tests, and a singular null collapse shows why invertibility is necessary. Conformal factors multiply under composition. For any symmetric B, the same exact decoder gives

$$\max_{i,j} |B_{ij} - (B_{00}/G_{00})G_{ij}| \le (1 - \sqrt{5}/5) \max_p |k_p^T B k_p|.$$

This bounds a metric residual, not distance to a Lorentz transformation. The prose theorem and exact controls are in `model/native_lorentz_transition` [8]. It supplies no source-derived chart transition or physical-null interpretation.

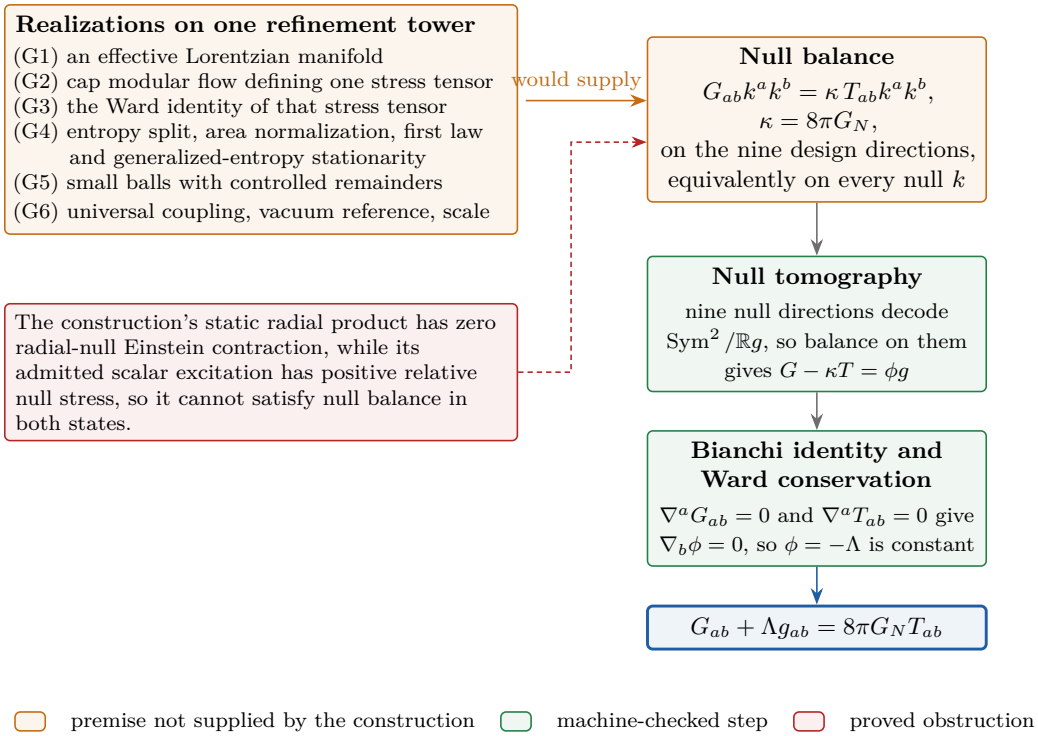


Figure 1: The Einstein equation as tensor completion of a realized null balance. Balance on the nine design directions is equivalent to balance on every null vector, and the tensor step is machine-checked from that premise. Realizations (G1) to (G6) on one refinement tower would supply it; the executable construction's static radial product fails it for its own scalar excitation.

Einstein shape. The minimal source-direction Einstein-shape theorem places supplied symmetric geometry and stress fields on the same finite event type. It asks for balance on nine fixed

algebraic source directions, four discrete step maps, Ward/Bianchi identities and step connectedness. Exact tomography derives all-null balance, the pointwise metric ambiguity, its constancy, and

$$\mathsf{geometry}_{ab} = \kappa\, \mathsf{stress}_{ab} + \Lambda \eta_{ab}.$$

The nine-set is algebraic, not selected by provenance or invariant as a set under Lorentz transformations. Identifying geometry with smooth curvature and stress with physical stress-energy requires the source-causal continuum, count–volume, stress, small-ball, coupling, scale and remainder receipts on one family, together with either a tensor-curvature reconstruction converging to the Einstein tensor or the independent continuum small-ball/null-balance identification. Scalar-curvature convergence alone is diagnostic. In the machine-checked continuum composition the premise is the null-contracted Einstein relation itself, or nine of its null components; the composition adds the constancy of the metric residue, the vacuum reference fixing Λ and the $8\pi G$ label. The small-ball step is Jacobson's entanglement-equilibrium argument, each of whose inputs enters as a receipt.

The modular route requires both a geometric cap-flow comparison and a compatible algebra-state limit. Finite support incidence alone supplies neither the state nor the geometric 2π normalization. The selected comparison must identify the same cap charges, stress tensor and clock throughout refinement. For a faithful Gibbs reference of a declared finite Hamiltonian one has the exact identity

$$S(\rho) - S(\tau) = \beta \operatorname{Tr}((\rho - \tau)H) - D(\rho\|\tau), \qquad \tau = Z^{-1} e^{-\beta H}.$$

This supplies a finite entropy–energy relation. It does not select β, identify a reduced regional state with the Gibbs state of its bare local Hamiltonian, or supply geometric entropy stationarity.

The executable construction. On a declared radial product over its refining spherical source, the native wave action of the executable construction has coefficient density $C = \mu \operatorname{diag}(-1, a)$, and $g = \sqrt{-\det C}\, C^{-1}$ has signature $(-,+,+,+)$ almost everywhere. Smoothing the angular variables converges in the norm-resolvent sense, so every fixed finite menu of observer readouts is approximated by one smooth static $(1+3)$-dimensional Lorentzian metric carrying a free scalar field [1]. The axial counter read through the source frame has a three-dimensional Brownian limit with radial generator $\frac{2}{99}(\partial_r^2 + \frac{2}{r}\partial_r)$, but complete angular histories are identical across distinct radial extensions, so neither that radius nor present observations select the radial product. This description asserts no exact smooth native metric, no responsive geometry and no physical stress, and discharges none of (G2)–(G6). It is the static geometry that fails null balance for its own scalar excitation.

Matching the field principal cone. The causal order and the field must share a clock and spatial dictionary. For a positive spatial principal tensor a, the static field symbol is $-\omega^2 + \xi^T a \xi$; its dual tangent form is $-dt^2 + dx^T a^{-1} dx$. Supply an invertible readback Jacobian $dx = J\, dz$ and a positive clock conversion $dt = \alpha\, d\tau$. The twelve null tests above are equivalent to

$$K = \operatorname{diag}(-\alpha^2, J^T a^{-1} J) = \frac{\alpha^2}{R^2} G, \qquad a = \frac{R^2}{\alpha^2} J J^T.$$

This criterion is nonvacuous: for any positive a, the pointwise choice $J = (\alpha/R) a^{1/2}$ works. It does not show that this Jacobian is the observer's readback or that a varying frame integrates to a global chart. A common invertible frame change preserves the equality by congruence; transforming only one side changes the proposed identification.

The unwarped radial product illustrates the distinction. In its round control $A = \kappa(-\Delta_{S^2})$, the fixed map $x = rn$ gives spatial tangent form

$$H_r = nn^T + (\kappa r^2)^{-1}(I - nn^T).$$

One global clock scaling matches flat radial and tangential cones only if $\alpha^2 = R^2$ and $\kappa r^2 = 1$. Distinct positive radii cannot both satisfy this; exact controls at radii one and two retain the mismatch. This excludes that fixed polar identification, not the product's curved Lorentzian description, other coordinates or local orthonormal frames. No radial action is replaced. The algebraic criterion and controls are in `model/field_cone_compatibility` [8]. Applying them requires event placements and the same field's actual clock and readback from one source. Principal-cone agreement alone fixes neither lower-order dynamics nor masses: $-\omega^2 + \xi^T a\xi + m^2$ has the same principal cone for every $m^2 \geq 0$.

Kinematic and Newtonian precursors are finite identities. On the declared Hermitian Lorentz module, $p = m\,u$ has Lorentz square m^2 and $p_0^2 = m^2 + |\mathbf{p}|^2$ in every oriented chart; no coordinate is identified with a physical energy. For a supplied integer layering, steady source law, shell equidistribution and shell count cn^2 through a finite depth, exact divergence bookkeeping gives outward flux $Q/(cn^2)$ per shell edge, while on a closed finite carrier an n^2 shell count at every positive layer forces $c = 0$, so the depth cutoff is explicit.

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1 A common matter metric and its gravitational response

This section is the interface of the executed construction with the gravitational layer of the tower. The conditional scalar description supplies a concrete dictionary for comparing matter and geometry, and the same dictionary yields the static-product control that any responsive geometry must overcome. The counts, ranks and identities quoted here are retained in the receipts of the matter and common-field contracts of the public repository. Write its four-dimensional coefficient density and metric as

$$C = \mu \operatorname{diag}(-1, 1, a), \qquad g = \sqrt{-\det C}\, C^{-1} = N^2 \operatorname{diag}(-1, 1, a^{-1}), \qquad N^2 = \mu\sqrt{\det a},$$

where a is a positive two-dimensional screen tensor and $\mu > 0$. These identities use the complete action density, including its measure. For the measurable native limit, smooth representatives are understood with the stated finite-menu approximation theorem and smearing transport.

1.1 Common principal symbols and finite field operators

Choose an orthonormal coframe e with $g = e^t \eta e$. Transporting Clifford matrices through its inverse gives

$$\{\gamma^\mu, \gamma^\nu\} = 2g^{\mu\nu} I, \qquad (\gamma^\mu p_\mu)^2 = (g^{\mu\nu} p_\mu p_\nu) I.$$

The scalar wave symbol, both Weyl principal determinants and the Maxwell symbol therefore have the same characteristic cone. The Maxwell gauge direction is retained, leaving two physical polarizations on a nonzero null covector. A change of clock coordinate rescales all field generators and the reference inverse temperature together, preserving βH. This pointwise dictionary does not select the fields' dynamics or chirality.

For a finite incidence factor C_0 on the declared angular–radial product,

$$D = \begin{pmatrix} 0 & C_0^\dagger \\ C_0 & 0 \end{pmatrix}, \qquad D^2 = \operatorname{diag}(C_0^\dagger C_0, C_0 C_0^\dagger).$$

The node block is the scalar spatial operator $B = C_0^\dagger C_0$. CAR second quantization gives a conditional fermion construction with node propagation $\cos(t\sqrt{B})$ and node-to-edge propagation $-iC_0 \sin(t\sqrt{B})/\sqrt{B}$. The full finite construction retains its extra edge modes. A factor matching the finite angular frequency is also tested; its ideal joint generator is not the exact Floquet generator of the finite angular/radial split word. Their difference retains the specified approximation budget.

Completing the actual tetrahedral cone through faces and tetrahedra gives a separate full-cochain control. Its dimensions are $(13, 42, 50, 20)$, its differential ranks are $(12, 30, 20)$, and its Betti numbers are $(1, 0, 0, 0)$. The complete metric $d + d^\dagger$ construction removes the truncated cone's thirty edge-cycle zero modes, preserves its scalar action block, and retains the scalar constant. This cone is a different global domain from the angular–radial cylinder. Differential-form parity does not establish physical Weyl chirality.

The smooth product admits a spin structure, but scalar radial Dirichlet conditions do not by themselves define a self-adjoint first-order spinor operator. A tested reflecting extension squares to a Neumann component and a Dirichlet component. Spin-connection and curvature terms also enter: a curved spinor square is not a componentwise scalar Laplacian. These domain and connection hypotheses are explicit in the common-field tests.

1.2 Area, volume and the entropy variation

The same product metric gives exact induced measures

$$dA = \mu\, d^2x, \qquad dV = \mu N\, dr\, d^2x.$$

Thus the scalar action measure supplies a whole-screen area dictionary. Fixed counting measure alone does not fix physical three-volume. Radial slices in this product have equal area and are not concentric small geodesic spheres.

Native constrained maximum-entropy completion and its joint counting first law hold on their finite algebraic variation domains. They do not identify fixed cap marginal or fixed total dimension with fixed proper volume. In the exact selected-reference test, forty-five four-port caps have identical counted edge entropy and crossing-seam counts ranging from ten to eighteen. No nonzero universal constant equates that entropy to this boundary proxy. The statement concerns a finite two-dimensional boundary proxy, rather than the whole-screen area above.

The family

$$\mu(z) = \mu_0 e^{-2z/3}, \qquad a(z) = e^{2z} a_0$$

keeps proper volume fixed on the declared radial coordinate interval and decreases its boundary area. At fixed model-coordinate inverse temperature, the cavity thermal entropy also decreases. Hence the trial functional $S_{\text{thermal}} + A/(4G_N)$ is not stationary for positive G_N on these variations. This is a specified thermal-cavity comparison; it is not an identification with local vacuum entanglement or a proof that repair generates those variations.

The distinction between observable algebras matters as well. For an executed symmetric mixture of $\pm\sqrt{\epsilon}$ displacements, the unread work density has positive first-order entropy and energy derivatives at fixed reference Hamiltonian. The retained sign-plus-work density has zero entropy derivative. Both outcomes and the purification are retained. A gravitational entropy identification must specify which region algebra and which records enter its bulk term.

1.3 A fixed-product obstruction and a necessary dynamic response

Proposition 1.1 (Radial null response of the static product)**.** *For a smooth metric*

$$g = f(x,y)\,\mathrm{diag}(-1, 1, h(x,y)), \qquad f > 0,$$

with positive screen metric h, both radial null contractions of its Einstein tensor vanish. With the Gaussian reference and other sources fixed, the same geometry cannot satisfy the Einstein balances for that reference and every coherent scalar excitation at positive gravitational coupling.

Proof. The (t, r) factor has Lorentz symmetry and all metric coefficients are independent of t, r. Its Ricci components obey $R_{tt} + R_{rr} = 0$ and $R_{tr} = 0$, also verified by direct differentiation for arbitrary smooth f, h. For $k = \partial_t \pm \partial_r$, $g(k,k) = 0$, so trace and cosmological metric terms vanish and $G(k,k) = 0$. The restored angular-constant scalar sector admits

$$\phi = A \sin(\pi r/L)\cos(\pi t/L).$$

It obeys the product wave equation and the declared radial Dirichlet condition. Its action-derived coherent relative stress satisfies

$$\Delta T(k,k) = (k\phi)^2 = \frac{A^2\pi^2}{2L^2} > 0 \quad \text{at } t = 0,\ r = L/4.$$

Subtracting the reference and excited Einstein equations contradicts $\Delta G(k,k) = 0$. This uses relative stress and supplies no absolute vacuum-energy prescription. □

A coherent amplitude changes stress quadratically; this argument is not a claimed failure of the linear entropy first law. A symmetric mixture of opposite displacements supplies a first-order state direction, with its observable algebra specified separately. Independently changing compensating sources and geometries outside the static product class are outside the proposition.

For the smooth spherical extension

$$ds^2 = -2du\,dv + R(u,v)^2 d\Omega^2,$$

direct curvature calculation gives $G_{uu} = -2R_{uu}/R$. The null Einstein balance consequently requires

$$R_{uu} = -4\pi G_N R T_{uu}, \qquad \theta_u = -\frac{\theta^2}{2} - 8\pi G_N T_{uu}, \qquad \theta = \partial_u \log(4\pi R^2).$$

Here u is affine along the declared null curves. These are necessary area-response equations in this symmetry class, not a selected repair law or a solution of the full Einstein system. The alternative finite nine-null theorem also requires its independent balances; a pointwise tetrad map alone does not supply them.

The exact block, null-stress and focusing algebra is formalized in the Lean library of the public repository, and the source matrices, curvature identities, complete finite records and approximation controls are executed with the observer-particle model. Their conditional realization establishes neither observed Standard Model masses nor generalized-entropy stationarity on a physical refinement family.

1.4 Matter action, regional entropy and the geometric equation

A finite particle spectrum and a responsive geometric readout do not by themselves specify the stress or entropy of a common spacetime theory. The curvature calculation makes this distinction explicit. In signature $(-,+,+,+)$ and units $\hbar = c = 1$, use the same complex Higgs doublet H as in the conditional matter action, write $X = H^\dagger H$, and specify

$$S = \int \sqrt{-g}\,[F(X)R + \mathcal{L}_{\min}], \qquad F(X) = F_0 - \xi X, \quad F_0 > 0. \tag{1}$$

Here $\mathcal{L}_{\min}$ contains the canonical gauge, Higgs, fermion and Yukawa terms. The covariant extension, metric, spin structure and curvature coupling ξ are supplied data. Let $T_{\min}$ be the Hilbert stress of this same minimally coupled action. Metric variation gives

$$\mathcal{E}_{ab} := 2FG_{ab} + 2(g_{ab}\Box - \nabla_a\nabla_b)F = T_{\min,ab}. \tag{2}$$

Equivalently, the additional matter stress relative to $2F_0G$ is $T_{\xi,ab} = 2\xi[XG_{ab} + (g_{ab}\Box - \nabla_a\nabla_b)X]$. It obeys $\nabla^a T_{\xi,ab} = -\xi R\nabla_b X$; this sector is not separately conserved. Total classical conservation uses all matter field equations, retaining gauge forces and Yukawa exchange. It does not establish a regulated quantum Ward identity.

The same curvature derivative $\partial(FR)/\partial R_{abcd} = (F/2)(g^{ac}g^{bd} - g^{ad}g^{bc})$ gives the Wald functional and its area coefficient,

$$S_W = 4\pi \int_{\partial B} F\,dA, \qquad \eta(X) = 4\pi F(X). \tag{3}$$

If $F_0 = 1/(16\pi G_0)$, then $\eta = 1/(4G_0) - 4\pi\xi X$. Positivity of F is a domain condition; a varying Higgs norm does not give a constant area coefficient. The Noether-charge functional in (3) is not, by definition, the entropy of the finite source state.

Proposition 1.2 (A controlled regional first variation)**.** *Vary about flat spacetime and constant $F = F_* > 0$, with variation parameter s and $f = \partial_s F|_0$. On the slice orthogonal to a unit timelike vector u, use radial normal coordinates and spatial metric tangent $h_{ij} = \partial_s g_{ij}|_0$, with $h(0) = \partial h(0) = 0$ and $h_{ij}(x)x^j = 0$. Define*

$$W = \int_B \frac{3F - F_*}{2F_*}\, dV, \qquad C_\ell = \frac{4\pi\ell^4}{15}.$$

If fourth directional derivatives of $\operatorname{tr} h$ and f on B_ℓ are bounded by M_h, M_f for every Euclidean unit direction, then

$$\delta S_W - \frac{8\pi F_*}{\ell}\delta W = -2\pi C_\ell(2F_*\delta G_{uu} - 2\Delta_3 f) + \mathcal{R}_\ell, \tag{4}$$

$$|\mathcal{R}_\ell| \le \pi^2\ell^6\left(\frac{3}{7}F_* M_h + \frac{20}{21}M_f\right). \tag{5}$$

All variations are first derivatives at $s = 0$; curvature and Laplacian values on the right are evaluated at the ball centre.

Proof. The curvature derivative contracts with the slice normal and spatial projector to $3F/2$, with reference coefficient $F_*/2$; its Noether generalized volume is therefore W above. Also $\delta W = \delta V + 3\int_B f/(2F_*)$ and $\delta S_W = 4\pi(F_*\delta A + \int_{\partial B} f)$. Spherical moments give the constant and quadratic scalar terms $4\pi\ell^2 f(0) + (2\pi/3)\ell^4\Delta_3 f$ on the boundary and $(4\pi/3)\ell^3 f(0) + (2\pi/15)\ell^5\Delta_3 f$ in the ball. The constant term cancels. The identity $\Delta_3 \operatorname{tr} h(0) = -(4/3)\delta G_{uu}(0)$ gives the metric term. Integrating the Taylor remainders $M|x|^4/24$ yields (5). □

The scalar variation is constrained by the Higgs coupling. For $\xi \neq 0$ and reference $H = (0, w)$ with real $w \neq 0$, the path $H_s = (0, w - sf/(2\xi w))$ realizes $\delta F = f$; for $\xi = 0$ it requires $f = 0$. Ordinary fixed proper volume instead leaves $16\pi^2\ell^2 f(0) - 2\pi C_\ell(2F_*\delta G_{uu} - 5\Delta_3 f) + O(\ell^6)$. Even $f(0) = 0$ does not remove the coefficient mismatch. Fixed generalized volume is a distinct admissibility premise. The Wald construction and generalized volume follow their Noether-charge and regional first-law formulations; the exact action, tensor and moment checks are in the repository [8].

For a geometric inference, the same state and action must supply $\delta S_{\text{bulk}} = 2\pi C_\ell \delta T_{\min,uu} + e_{\text{bulk}}$ and fixed-W stationarity with error e_{stat}. Then

$$|\delta(\mathcal{E}_{uu} - T_{\min,uu})| \le \frac{|e_{\text{bulk}}| + |e_{\text{stat}}| + |\mathcal{R}_\ell|}{2\pi C_\ell}. \tag{6}$$

Ten timelike probes spanning the symmetric tensors determine the full residual, provided they use common metric, stress, state and clock data. Nonminimal boundary counterterms must be assigned consistently between bulk entropy and S_W. With $f = 0$ this is the constant-coefficient linearized Einstein normalization; finite nonlinear regional deficits and errors $o(\ell^4)$ are required for a nonlinear inference. A common isotropic vacuum term becomes a spacetime constant only with constant gravitational coupling and the matched total Ward and Bianchi identities.

The finite winding spectra with centered contact provide interacting energy levels on prescribed textures. Their radial conserved current is eta-pairing, not an established Standard Model charge map. The separate native-current matter carrier has a specified family architecture, and its stationary flavor coefficient uses a declared static mean-feedback service. Summing every memory lag of that service gives a completely positive corrected reduced semigroup on stationary product preparations and unread outputs, with a coherent ordering potential, positive diffusion and an explicit finite-time error of second order in the pulse step [2]. An explicit carrier extension admits

a localized CAR inclusion intertwining the native currents and a pixel-dependent quadratic mass operator. This finite inclusion does not supply an interacting continuum vacuum or its gravitating stress. The ordered-spin texture energy and the fermion occupation energy are distinct terms; a physical rest-energy assignment requires their joint variation relative to the same vacuum. An interacting continuum join must transport one source's field algebra, currents, stress and state into the regional data of (6), with a common proper clock and controlled errors. The finite algebraic and spectral results supply no such physical entropy identification or common continuum model.

The compact opposite-degree texture has finite energy bounds and a nonzero localized spatial operator. The quartic construction of the particle paper [2] retains that operator on the same band, with exact slice spectra, full-core vacuum bounds and Gibbs response. The separate pixel feedback consumes quadratic mass coefficients; its replacement by the interacting spectral measure and a physical vacuum identification require additional maps.

2 The modular input to gravitational response

Theorem 2.1 (Conditional cap modular normalization)**.** *Let one cofinal refinement tower have a common support-visible geometric cap algebra and state in its mixed-GNS limit. Suppose its finite collars have exact Markov replacements or controlled faithful Markov comparison, the local algebra-state comparisons and replacements are compatible under refinement, and modular action is readable as a map of cap supports. Suppose further that a cap-normal certificate supplies oriented cross-ratio rigidity and an independent geometric 2π-KMS comparison. Then the limiting modular automorphism of each geometric cap is its cap-preserving conformal flow with the 2π normalization. This is an automorphism statement on the limit algebra; a literal $K_C = 2\pi B_C + Z_C$ representation requires an additional type-I realization, with Z_C central.*

Proof outline. Controlled Markov replacement confines the finite modular defect to the collar, with the faithful-collar comparison controlling its matrix elements on fixed local observables. Projective compatibility and mixed-GNS convergence define one limiting cap pair independently of a chosen comparison sequence. Support readability turns its modular action into a cap-support flow. Oriented cross-ratio rigidity identifies this flow with the cap-preserving conformal one up to parameter scale; the independent KMS comparison fixes that scale to 2π. None of these steps identifies the full algebra with a type-I factor or derives the certificate from the unchanged native repair law. □

The corresponding half-sided null inclusion has a positive translation generator P_Ω and, on its common quadratic-form core, the affine relation $K_a(\Omega) = K_0(\Omega) - 2\pi a P_\Omega$. Reading its endpoint derivative as a local charge $\int T_{kk}$ requires effective locality and an identification with the *same* stress tensor that propagates the matter fields. Affine/projective interval transport then produces the ball kernel used below, provided the endpoint and collar remainders meet the displayed small-ball bound. Thus cap modular covariance, the local stress reading and the entropy–area response are three separately testable inputs; the finite record entropy identity by itself supplies none of their physical identifications.

3 Null balance and the Einstein tensor

The tensor theorem isolates the precise content of the gravitational conclusion. A successful source realization must produce its balance premise independently from matter, geometry and entropy.

Theorem 3.1 (Null tomography)**.**

Let V be four-dimensional with Lorentz form $\eta = \mathrm{diag}(-1,1,1,1)$. A symmetric form X vanishes on every null square exactly when $X = \phi\eta$. Nine null directions suffice to recover X modulo this line: the six $(1, \pm e_i)$ and $(1,s,s,s), (1,s,s,-s), (1,s,-s,s)$, where $s = 1/\sqrt{3}$. The corresponding exact decoder has $\|x\|_\infty \le (2+\sqrt{3})\|q\|_\infty$ in its registered coordinates.

Proof. On the unit sphere, $X(1,n;1,n) = X_{00} + 2X_{0i}n_i + X_{ij}n_in_j$. Opposite directions remove the mixed temporal entries. The six axis values fix the spatial diagonal modulo the metric, and the three remaining sign choices fix the spatial off-diagonal entries. Thus the kernel consists exactly of the metric line. Solving this finite system gives the stated decoder bound; its determinant and identities are also checked in the finite theorem library [8]. □

Theorem 3.2 (Einstein equation from a common null balance)**.** *Let (M,g) be a connected time-oriented C^3 Lorentzian four-manifold and T a symmetric C^1 tensor with $\nabla^a T_{ab} = 0$. Fix a single $\kappa = 8\pi G_N > 0$. If*

$$G_{ab}k^ak^b = \kappa T_{ab}k^ak^b$$

for every null k at every point, then there is a constant Λ such that $G_{ab} + \Lambda g_{ab} = \kappa T_{ab}$. Balance on the nine directions in any orthonormal frame suffices.

Proof. Null tomography gives $G - \kappa T = \phi g$ pointwise. The contracted Bianchi identity, the independently supplied Ward identity and metric compatibility give $\nabla\phi = 0$. Connectedness makes ϕ constant; set $\Lambda = -\phi$. Its value requires one reference condition. □

3.1 The small-ball input and its normalization

The missing physical step in the preceding theorem is its null-balance premise. Under (G1)–(G6), a sufficiently small cap has a tangent causal diamond D_ℓ. In local inertial coordinates its preserving conformal Killing field and its lapse on the central slice are

$$\xi_{D_\ell} = \frac{(\ell^2 - r^2 - t^2)\partial_t - 2tx^i\partial_i}{2\ell}, \qquad \xi^t_{D_\ell}\big|_{t=0} = \frac{\ell^2 - r^2}{2\ell}.$$

The geometrically normalized cap modular flow and the same-source local stress reading in (G2) give the ball modular first law

$$\delta S_{\mathrm{bulk}} = 2\pi \int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\delta T_{uu}\, d^3x + \delta E_{\ell,\delta}.$$

Here $\delta E_{\ell,\delta}$ carries finite-collar, modular-transport and locality errors. The factor 2π is a geometric KMS normalization required by (G2), not a consequence of entropy maximization. Direct integration gives

$$\int_{B_\ell} \frac{\ell^2 - r^2}{2\ell}\, d^3x = \frac{4\pi\ell^4}{15}.$$

Thus, for stress smooth on the ball,

$$\delta S_{\mathrm{bulk}} = \frac{8\pi^2\ell^4}{15}\,\delta T_{uu} + \delta E_{\ell,\delta} + O(\ell^5\|\partial T\|).$$

The independent fixed-proper-volume geometric identity required by (G4) is

$$\delta A\big|_V = -\frac{4\pi\ell^4}{15}\,\delta G_{uu} + \epsilon_A.$$

With fixed-volume stationarity $\delta S_{\rm bulk} + \delta A/(4G_N) = \epsilon_{\rm stat}$, the two coefficients give the finite-scale relation

$$\delta G_{uu} - 8\pi G_N \delta T_{uu} = \frac{15 G_N}{\pi \ell^4} \left(\delta E_{\ell,\delta} + \frac{\epsilon_A}{4G_N} - \epsilon_{\rm stat} \right) + O(\ell \|\partial T\|).$$

This displays the required cutoff control: every entropy, area and stationarity remainder in parentheses must be $o(\ell^4)$ along the same refinement and small-ball limit. For example, if a fixed-collar comparison bounds $|\delta E_{\ell,\delta}|$ uniformly over the chosen ball family by a constant times the constructive recovery error, the exact-Markov replacement error and a regularized modular-transport error, its first two terms may be written $r_{\rm FR}(\varepsilon) = 2\sqrt{1 - e^{-\varepsilon}} \le 2\sqrt{\varepsilon}$ and $4\lambda_*^{-1} d_{\rm M}(\varepsilon)$, where $\lambda_* > 0$ is the fixed faithful-collar spectral floor and $d_{\rm M}$ bounds distance to an exact Markov state. Their sum with the transport error must be $o(\ell^4)$ on the chosen joint limit. A bound that is merely $o(1)$ at fixed ℓ cannot be divided by ℓ^4 without specifying an order of limits or a diagonal sequence. Exact Einstein dynamics is a controlled-limit statement, not a finite-cutoff identity.

If this relation holds for every local timelike unit vector in matched orthonormal frames, $g(u,u) = -1$ keeps the cosmological term fixed under the scalar variation. For $u = \gamma(1, v^i)$, the polynomial $\delta Y_{00} + 2v^i \delta Y_{0i} + v^i v^j \delta Y_{ij}$ vanishes on the open unit ball, where $Y = G + \Lambda g - 8\pi G_N T$ and the common reference fixes Λ. Its coefficients vanish, so $\delta Y = 0$. A common maximally symmetric reference, compatible variations on the connected branch, and the independent Ward identity complete the tensor equation. Equivalently, once the same-source construction yields the nine null balances, the preceding null-tomography theorem gives $G + \Lambda g = 8\pi G_N T$ directly. These are the conditional entanglement-equilibrium inputs of [9]; finite entropy maximization alone supplies none of the geometric area identity, universal coupling or same-source stress reading.

The coefficient G_N is an area-to-entropy conversion on this branch. If the shared-cut edge entropy density is $\bar{\ell}_{\rm shared}$ per cell of physical area $a_{\rm cell}$, its normalization is

$$G_{\rm geom} = \frac{a_{\rm cell}}{4\bar{\ell}_{\rm shared}}.$$

The conditional pixel and scale identifications $a_{\rm cell} = P\ell_*^2$ and $\bar{\ell}_{\rm shared} = P/4$ then give $G_{\rm geom} = \ell_*^2$ and $G_{\rm SI} = c^3 \ell_*^2/\hbar$. The cancellation of P is exact; the shared-cut density and the independent physical scale certificate require realization on the same gravitational tower. Local null data leave Λ invisible, so its value requires a reference or global capacity condition rather than this small-ball calculation.

For nonminimal $F(H^\dagger H)R$ coupling, the correct curvature entropy and fixed generalized volume are instead those in Section 1.4. Replacing them by constant area and ordinary fixed volume changes the field equation. Every remainder must be $o(\ell^4)$ in the same small-ball limit; a fixed absolute error does not suffice after division by ℓ^4.

3.2 Controlled matter inputs and the common-model boundary

The centered lattice Weyl source has a controlled free limit on $0 \le U = o(c^{2/7})$, with $c = v/a \to \infty$. On the exactly free branch $U = 0$, all eight valleys remain. The resulting positive-energy free field has a conserved Lorentz stress tensor and geometric double-cone modular flow. Compact lattice state perturbations and finite quadratic detectors converge to its leading stress response in a joint spacing/small-ball limit. Nine explicit shell perturbations span traceless stress variations; the finite boosted-ball response has full rank for $0 < R/L \le 2^{-28}$. On each fixed ball, the finite-density modular flows converge on fixed probes, with a generator graph core and convergent selected relative entropies. No uniform lattice covariance gap is assumed. The source and proofs are in the repository [8].

A separate local electric completion concerns a changing Euclidean overlap projector $P(\theta)$, rather than a space-time metric. The connection $C = [P, P']$ satisfies $P' + [C, P] = 0$. For an orthonormal frame V of its range,

$$V^\dagger(-\partial_\theta^2)V = -(\partial_\theta + V^\dagger V')^2 + V'^\dagger(I - P)V'.$$

The final positive term is the differential Born–Huang scalar. A finite parent $K+\Lambda Q$, $K \succeq 0$, retains the complementary states; if a_k is the kth eigenvalue of PKP and $\Lambda > a_k$, its first k eigenvalues lie within $\|PKQ\|^2/(\Lambda - a_k)$ below the compressed eigenvalues. The complete 576-component control verifies this bound. Its coordinate, boundary and kinetic law are supplied. Separate free overlap propagation and a 438/1596-state interacting gauge–Higgs response supply no common interacting stress tensor or responsive metric [2].

A separate fixed-spacing spin construction is interacting and admits ordered gapless continuum observables; a surviving connected interaction and sharp relativistic continuum mass shell are not established. The compact quartic construction of the particle paper [2] supplies exact finite charged slice poles and one interacting localized Hamiltonian, whose mean form $\bar{H}_W$ on 180 localized annihilators contains the band kinetic term, the memory pairing and the quartic $\lambda(N_W - 90)^2$, $\lambda \geq 0$. It does not turn the free Weyl stress theorem into an interacting stress theorem or provide a physical metric. Gravity needs two of its finite identities. For the declared faithful Gibbs reference $\rho_\beta = Z^{-1}e^{-\beta\bar{H}_W}$, $\beta > 0$, and $H(s) = \bar{H}_W + sX$, with X not assumed to commute with $\bar{H}_W$,

$$\rho'_\beta(0) = -\beta \int_0^1 \rho_\beta^u (X - \langle X\rangle)\rho_\beta^{1-u}\, du, \qquad S' = \beta(E' - \langle X\rangle).$$

A hopping control $h_s = sT + MV$, with $1/2 \leq s \leq 3/2$, $M \geq 8$ and $\|T\| < 2$, moves the four-dimensional band, with projector P_s and $N_s = \mathrm{d}\Gamma(P_s \otimes I_{45})$. On a complete memory record ω its exact work operator is

$$\partial_s H_{\omega,s} = \mathrm{d}\Gamma(T \otimes I_{45}) + \lambda\{N_s - 90, \mathrm{d}\Gamma(P'_s \otimes I_{45})\},$$

and in a transported comparison frame W_s the intrinsic force $J_s = \mathrm{d}\Gamma((W_s^* T W_s) \otimes I_{45})$ obeys $\|J_s\| \leq 360$ uniformly in the exterior volume, equals that work operator in contrasts of even localized states with a common complement reference, and enters the Gibbs derivative $S' = \beta(E' - \mathrm{Tr}(\rho_s J_s))$. The hopping amplitude is a declared control, so this work derivative supplies no identification with a metric variation or conserved Lorentz stress, and these finite identities supply no geometric modular flow, area normalization or fixed-volume generalized-entropy stationarity.

Pressure in (G2) also needs a common energy and a strain law. Consensus about identical fixed-volume records cannot select their unobserved strain derivative: $H_\alpha(V) = Q(V/V_0)^{-\alpha}$ gives $p/\rho = \alpha$. Three geometrically spaced energy readings identify the response within the held-state family $H(V) = C + A(V/V_0)^{-\alpha}$, $A > 0$, $\alpha \geq 0$, and an additive energy offset changes p/ρ while leaving pressure unchanged [3].

These results close specified matter inputs on their own domains. Conditions (G1)–(G6) require a common event geometry, reference, physical clock, universal coupling, entropy–area normalization and stationarity under geometry changes. The static-product obstruction in Proposition 1.1 proves why merely adjoining a fixed radial geometry cannot perform this remaining step.

A Causal order, counts and clocks on source populations

The theorems of this appendix concern supplied source populations, read laws and model clocks. They prove the causal, count and clock limits under those premises; the native process must produce the population and law before the limits apply to it.

A.1 Causal order on a source-metric population

The continuous rank-three repair carrier admits finite populations chosen by its own metric. A covering property of such a population also supplies a constructive causal approximation, with an explicit local read law and model clock. Let Ω be a convex subset of a real normed space, and let $S \subset \Omega$ satisfy

$$\forall x \in \Omega \quad \exists s \in S : \quad \|s - x\| \le h, \qquad h \ge 0.$$

For an edge radius $a > 2h$, put an event at every (j, s), $j \in \{0, \dots, K\}$, $s \in S$. An event at $(j+1, t)$ reads every (j, s) with $\|t - s\| \le a$. This includes the same-site read, so waiting is permitted. The event order is the reflexive transitive closure of these reads. Reading every permitted neighbour, keeping the full population at each layer, and assigning a duration $\Delta = a/c$ with $c > 0$ are hypotheses of this construction. They are substantive choices of dynamics and clock.

Proposition A.1 (Covering constructs causal paths)**.** *For $s, t \in S$ and $k \ge 1$, with both layers present,*

$$\|t - s\| \le k(a - 2h) \quad \Longrightarrow \quad (j, s) \preceq (j + k, t) \quad \Longrightarrow \quad \|t - s\| \le ka. \tag{7}$$

If S is finite and nonempty, this layered poset has width exactly $|S|$. Every event at layer j has height j, counting edges from layer zero.

Proof. Divide the segment from s to t into k equal pieces, with division points $x_i = s + i(t - s)/k$. Convexity gives $x_i \in \Omega$. Keep the endpoints $p_0 = s$, $p_k = t$ and choose $p_i \in S$ with $\|p_i - x_i\| \le h$ at every interior division point. Then

$$\|p_{i+1} - p_i\| \le \|t - s\|/k + 2h \le a.$$

These are actual permitted reads. For any permitted path the triangle inequality gives the opposite bound in (7). A layer is an antichain, and the same-site waiting chains partition all events into $|S|$ chains. This proves the width statement. Every read increases the layer by one, and a waiting chain reaches every layer, proving the height statement. □

The path construction and outer displacement bound hold in arbitrary real normed spaces; `Lean/Geometry/SourceNetCausalCone.lean` formalizes them. The partial order, its equality with the reflexive transitive closure of the reads, the graph-distance characterization, and the width and height statements are formalized in `Lean/Geometry/SourceNetLayeredOrder.lean`. The volume estimates below concern the three-dimensional Euclidean repair carrier. The vertical weighted bound and its family limit are formalized below; the separate general moving-tip finite estimate is analytic.

For model time $\tau = k\Delta$, the two radii in (7) are $c(1 - 2h/a)\tau$ and $c\tau$. Consider a family with

$$a_n \longrightarrow 0, \qquad h_n/a_n \longrightarrow 0. \tag{8}$$

The read law agrees with the speed-c continuum causal order away from its null boundary. More explicitly, take two continuum events with positive time separation whose positions lie in Ω. Sample positions within h_n and times on the layer grid within Δ_n. A strict timelike margin eventually exceeds the endpoint errors, the time rounding errors, and $2ch_n/a_n$ times the sampled duration; the inner implication then applies. A strict spacelike margin eventually exceeds the endpoint and time errors, and the outer implication forbids a path. This statement requires the sampled layers to exist in the chosen time window. It makes no assertion of exact order reflection for null-separated events at a finite resolution.

Volume from a controlled population measure. Suppose $\Omega \subset \mathbb{R}^3$ is bounded and convex, and S is finite. In addition to the h-covering population S, supply a measurable partition $(C_s)_{s\in S}$ of Ω, up to null sets, with positive masses $v_s = \mathrm{vol}(C_s)$ and assignment error

$$y \in C_s \quad \Longrightarrow \quad \|y - s\| \le H \quad \text{for almost every } y. \tag{9}$$

Clipped Voronoi cells allow $H = h$. Equal-volume cells with a different assignment bound are also admitted; no Voronoi hypothesis is required. For $x \in S$ and $T = K\Delta > 0$, let I be the inclusive discrete interval between $(0, x)$ and (K, x), and assign it spacetime mass

$$V_{S,\Delta}(I) = \Delta \sum_{(j,s)\in I} v_s.$$

The spatial assignment rule and the temporal factor are part of the measure. The limiting volume convention is $dt\, d^3x$. For the metric $c^2dt^2 - |dx|^2$, its metric volume is c times this coordinate volume.

Proposition A.2 (Weighted Alexandrov-volume limit)**.** *Assume the preceding hypotheses and $B(x, cT/2 + H) \subset \Omega$. Then*

$$\left| V_{S,\Delta}(I) - \frac{\pi c^3 T^4}{24} \right| \le 4\pi(T + \Delta)(cT/2 + H)^2 \left(H + \frac{cTh}{a} \right) + \frac{\pi c^3 T^3 \Delta}{2}. \tag{10}$$

Thus $H_n \to 0$ and (8) imply the continuum Alexandrov-volume limit on aligned, bounded interior time windows. If all cells have a common mass v, this is the raw event-count statement $v\Delta\,|I| \to \pi c^3 T^4/24$.

Proof. Write $u_j = \min(j\Delta, T - j\Delta)$ and $r_j = cu_j$. The causal sandwich puts every population point within $c(1 - 2h/a)u_j$ of x in the interval, and puts every interval point within r_j. At the endpoint layers the interval contains precisely the single endpoint. The partition assignment therefore gives, at every layer,

$$\frac{4\pi}{3}\big(c(1 - 2h/a)u_j - H\big)_+^3 \le \sum_{(j,s)\in I} v_s \le \frac{4\pi}{3}(r_j + H)^3.$$

For the lower estimate, a point in the smaller ball is assigned to a population point inside the inner causal radius. For the upper estimate, the union of cells assigned to interval points lies in the larger ball. The stated interior buffer keeps these balls in the spatial window. The deviation of either radius from r_j is at most $H + cTh/a$. Since the derivative of $(4\pi/3)r^3$ is $4\pi r^2$, the error per layer is bounded by $4\pi(cT/2 + H)^2(H + cTh/a)$.

Finally, $F(t) = (4\pi/3)[c\min(t, T - t)]^3$ has Lipschitz constant $\pi c^3 T^2$, vanishes at both endpoints, and integrates to $\pi c^3 T^4/24$. Its left Riemann sum differs from its integral by at most $\pi c^3 T^3 \Delta/2$. The inclusive right endpoint contributes zero to this reference sum. Summing the spatial errors over $K + 1$ layers proves (10). □

The complete finite estimate, with almost-everywhere cell coverage, disjointness and assignment, is formalized by `SourceNetVolumeError.weighted_alexandrov_error` in `Lean/Geometry/SourceNetVolumeError.lean`. The zero-hop endpoint cases and all $K + 1$ layers are included. The general limit for varying, possibly unequal cell masses is proved by `SourceNetVolumeError.weighted_alexandrov_tendsto`: it assumes $a_n, H_n, h_n/a_n \to 0$, $K_n(a_n/c) \to T$ and the eventual interior buffer, and derives convergence of the actual weighted generated intervals. `Lean/Geometry/GoldenSourceVolumeLimit.lean` instantiates the bound with the actual golden sites and equal cell masses, proves its exact raw count interpretation, and derives the aligned weighted-volume limit as $K_n\Delta_n \to T$, under the stated interior buffer.

Proposition A.3 (Interior diamonds with moving tips)**.** *Let $x, y \in S$, $\|y - x\| < cT$, and $T = K\Delta > 0$. Suppose the continuum diamond between $(0, x)$ and (T, y) has a positive spatial buffer inside Ω. Write*

$$D(t) = B(x, ct) \cap B(y, c(T - t)), \qquad V(D) = \int_0^T \mathrm{vol}(D(t))\, dt.$$

For the inclusive discrete interval I between these tips,

$$|V_{S,\Delta}(I) - V(D)| \le 8\pi(T + \Delta)(cT + H)^2 \left(H + \frac{2cTh}{a}\right) + 4\pi c^3 T^3 \Delta. \tag{11}$$

In particular, the same refinement conditions give weighted interval-volume convergence for arbitrary fixed interior timelike diamonds.

Proof. At layer j, the union of assigned cells contains the intersection of the two balls with radii $(c(1 - 2h/a)j\Delta - H)_+$ and $(c(1 - 2h/a)(T - j\Delta) - H)_+$, up to null sets. It is contained in the intersection with radii $cj\Delta + H$ and $c(T - j\Delta) + H$. Zero-radius balls contribute zero volume. Changing either reference radius changes the intersection only within that ball's spherical shell. Each radial error is at most $H + 2cTh/a$, and each outer radius is at most $cT + H$. The sum of the two shell bounds gives the spatial term in (11). The reference section volume has Lipschitz constant at most $8\pi c^3 T^2$: over a time increment, each of the two radii changes at speed c, and each ball has surface area at most $4\pi c^2 T^2$. Both endpoint sections have zero volume. The same Riemann-sum argument gives the temporal term. □

A conservative-record family with a raw count law. The population and equal-mass hypotheses have an explicit realization in conservative source records. In the exact source-axis presentation, write $\varphi = (1 + \sqrt{5})/2$ and $L = 2/\sqrt{\varphi + 2}$. For consecutive Fibonacci numbers $q = F_n$, $p = F_{n+1}$, with $n \ge 4$, define

$$\xi_b = b\varphi - \lfloor b\varphi \rfloor, \qquad 0 \le b < q.$$

For a triple $b = (b_1, b_2, b_3)$, put $m_i = -\lfloor b_i \varphi \rfloor$ and take the integer record

$$z(b) = (b_2 - m_1, b_2 + m_1, b_3 - m_2, b_3 + m_2, b_1 - m_3, b_1 + m_3). \tag{12}$$

Its coordinate sum is $2(b_1 + b_2 + b_3)$. It therefore lies in the exact conservative seam-current image, whose integer section supplies a finite source word. Direct contraction with the source axes gives the readback

$$s(b) = L(\xi_{b_1}, \xi_{b_2}, \xi_{b_3}) \in \Omega = [0, L]^3.$$

These are actual source-record images. A regular mesh is used below only to prove a volume estimate; it does not replace the event positions.

Proposition A.4 (Source-record quadrature and four-dimensional counting)**.** *The population $S_q = \{s(b) : 0 \le b_i < q\}$ has exactly q^3 distinct sites. It admits a partition of Ω into cells of common volume $v_q = L^3/q^3$, assigned to the sites with error*

$$H_q = \frac{2\sqrt{3}L}{q}. \tag{13}$$

Consequently, for every Lipschitz function f on Ω,

$$\left| v_q \sum_{s \in S_q} f(s) - \int_\Omega f(x)\, d^3x \right| \le L^3 H_q \,\mathrm{Lip}(f).$$

Apply the preceding complete-neighbour read law on S_q, with $a_q = L/\sqrt{q}$ and $\Delta_q = a_q/c$. Its density in the declared $dt\, d^3x$ convention is

$$\rho_q = \frac{q^3}{L^3 \Delta_q}. \tag{14}$$

Normalized raw event counts converge to volume on fixed bounded regions with volume-null boundary. On intervals whose endpoints converge to a strictly timelike pair and whose limiting diamond has an interior buffer in the spatial and temporal windows, the generated order intervals converge to that same diamond in count measure. In particular, for vertical tips separated by model time T,

$$\frac{|I_q|}{\rho_q} \longrightarrow \frac{\pi c^3 T^4}{24}.$$

The local limiting order and measure are those of flat $1+3$ spacetime. This conclusion concerns this declared source-record family and update law.

Proof. The Fibonacci recurrence gives $\gcd(p,q) = 1$ and $p - q\varphi = (-1)^n \varphi^{-n}$, hence $|\varphi - p/q| < q^{-2}$. Let k_b be the residue of bp modulo q. These residues permute $0, \dots, q-1$. For $b = 0$ the two floors below are zero. For $0 < b < q$, bp/q is at least $1/q$ from either adjacent integer, whereas $|b(\varphi - p/q)| < 1/q$. Therefore

$$\lfloor b\varphi \rfloor = \lfloor bp/q \rfloor, \qquad \xi_b = \frac{k_b}{q} + b(\varphi - p/q), \qquad \left| \xi_b - \frac{k_b}{q} \right| < \frac{1}{q}.$$

Irrationality of φ makes the orbit points distinct; the source readback then distinguishes all q^3 records.

Assign to $s(b)$ the rectangular cell $\prod_i [Lk_{b_i}/q, L(k_{b_i}+1)/q]$, resolving cell-boundary ties arbitrarily. This is a partition up to null sets. A site need not belong to its assigned cell, but every coordinate displacement from a point of that cell is at most $2L/q$. This proves (13), the covering property and the Lipschitz estimate by integrating the displacement bound over each cell. Taking products with time cells $[j\Delta_q, (j+1)\Delta_q]$ proves convergence of the normalized event measure: for a Lipschitz test function on an aligned time window of length T, the error is at most

$$TL^3 (H_q \operatorname{Lip}_x(f) + \Delta_q \operatorname{Lip}_t(f)).$$

An extra endpoint layer contributes at most $L^3 \Delta_q \|f\|_\infty$. Continuous approximation then gives the count statement for regions with volume-null boundary.

Use $h_q \le H_q$ in the causal sandwich. Here $a_q \to 0$ and $H_q/a_q = 2\sqrt{3}/\sqrt{q} \to 0$; the sufficient positive-inner-cone condition from this bound holds when $q > 48$. For converging timelike tips, the order and sampling discrepancies are confined to shrinking neighbourhoods of the two limiting null boundaries. Their volume is zero. The uniform cell displacement and temporal error also tend to zero, so the normalized number of events in these shrinking neighbourhoods tends to zero. This proves interval-count convergence; for vertical tips it also follows directly from (10). The formula for the flat diamond is its elementary ball-section integral. □

The spatial arithmetic, partition and quadrature statements are formalized in `Lean/Geometry/GoldenSourceAssignment.lean`; normalized event-count convergence on fixed measurable continuity sets for the declared time grid, with final-layer discrepancy at most $L^3\Delta_q$, in `Lean/Geometry/GoldenSourceCountLimit.lean`. Causal-frontier and pairwise null-separation nullity are proved by Fubini in `Lean/Geometry/SourceCausalBoundary.lean`, and eventual order agreement is derived from actual covering paths in `Lean/Geometry/SourceNetOrderLimit.lean`. Count

convergence for moving generated intervals is proved in `Lean/Geometry/GoldenSourceCausalLimit.lean`; its reference volume is restricted to the supplied time and spatial window, and an uncut diamond requires the containment hypothesis. Strict generated-pair convergence to the geometric product measure with the $N_q(N_q-1)$ normalization is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`, the explicit vertical volume error and the aligned weighted count-volume limit in `Lean/Geometry/GoldenSourceVolumeLimit.lean`, and the uncut interval volume together with the ordering-fraction limit of Proposition A.5, with the window containment hypothesis explicit, in `Lean/Geometry/FlatDiamondNormalization.lean`. The moving-tip finite error estimate (11) is an analytic estimate.

Proposition A.5 (The four-dimensional ordering fraction)**.** *For the contained timelike intervals in Proposition A.4, let $N_q = |I_q|$ and let C_q count their strict causally ordered pairs. Then*

$$\frac{2C_q}{N_q(N_q-1)} \longrightarrow \frac{1}{10},$$

the four-dimensional Myrheim–Meyer fraction [10, 11, 12].

Proof. The cell assignment gives convergence also for the product event measure. Interval-membership boundaries, the pairwise null-relation boundary and the coincidence diagonal have zero limiting measure. The causal sandwich confines disagreement to shrinking neighbourhoods of these boundaries. Hence the normalized strict-pair count tends to its continuum double integral; $N_q \to \infty$ removes the distinction between N_q^2 and $N_q(N_q-1)$.

In units $c = 1$, place the limiting tips at $(0,0)$ and $(T,0)$ and write $V = \pi T^4/24$. The volume to the future of (t,x) within the interval is $\pi[(T-t)^2 - |x|^2]^2/24$. Thus the strict oriented-pair integral is

$$\frac{\pi^2}{6}\int_0^T \int_0^{\min(t,T-t)} r^2\big((T-t)^2 - r^2\big)^2\, dr\, dt = \frac{\pi^2 T^8}{11520} = \frac{V^2}{20}.$$

The ordering fraction counts both orientations of distinct comparable pairs, giving $1/10$. Constant changes of time and volume units cancel in the ratio. This is an interval statistic; the same value is not asserted for a whole spatial box times a time window. □

The strict-pair count converges to the geometric double-indicator measure, with the $N_q(N_q-1)$ normalization retained. The product endpoint clipping is bounded by $L^6\Delta_q(2T+\Delta_q)$ in `Lean/Geometry/SourceCountTransport.lean`, and the pair limit is proved in `Lean/Geometry/GoldenSourcePairLimit.lean`.

The spherical-section, Lorentz-volume and pair-integral modules derive the geometric measure by Fubini and radial integration. Their explicit Lorentz boost has determinant one and preserves the causal relation. The normalization theorem in `Lean/Geometry/FlatDiamondNormalization.lean` gives the $1/10$ limit for supplied $c > 0$ and timelike limiting tips whose diamond lies in $[0,T]\times[0,L)^3$. Measure equality retains the null final-time boundary. These theorems select no population, read law, physical clock or physical measure, and they do not count additional instrument or routing operations.

A sharper finite covering constant is available without changing this construction. Sort the one-dimensional orbit as $0 = \eta_0 < \cdots < \eta_{q-1} < 1$ and put

$$r_q = \max\left\{\frac{1}{2}\max_{i<q-1}(\eta_{i+1}-\eta_i), 1-\eta_{q-1}\right\}, \qquad h_q = \sqrt{3}Lr_q.$$

The product population has exactly this covering radius in the closed cube. The path condition $2h_q < a_q$ is the exact quadratic-field inequality $12qr_q^2 < 1$. This sharper h_q concerns paths; the assignment estimate H_q in the count proof is separate.

The source-word section has coefficients growing at most linearly in q; the finite implementation records their actual primitive lengths. Counting one distinguished event per site and layer is part of the stated law. Intermediate seam-word operations, read instructions and bookkeeping records are not silently included in the same density. The spatial populations are nested along increasing Fibonacci cutoffs. Their changing layer duration and edge radius do not make the spacetime histories induced prefixes; regulator convergence uses the common metric carrier and controlled cell assignments.

Source populations and physical interpretation. The finite-word source construction supplies arbitrarily fine metric nets in bounded convex windows with nonempty interior in the rank-three completion. A maximal separated subset retains a controlled fill distance and three-dimensional packing bounds. These statements supply the spatial hypotheses of Proposition A.1 under a specified resolution and population rule. Three-dimensional packing bounds alone do not supply equal-volume sampling. The count conclusion requires a partition satisfying (9) and, for unweighted counts, equal cell masses or a separate counting approximation.

The construction gives a causal approximation on source-metric points without inserting a Cartesian spatial table. The complete neighbour read law, repeated population, cell measure, and model duration are supplied. An implementation by accepted observer repairs must realize these inputs with its actual records and operation costs. A scalar action on the same population can have a different finite propagation relation; the causal cone proved here is not an identification of that action's stencil cone or its continuum response speed. Agreement with a field action and a physical clock is an additional common-model condition.

A.2 Source-population quadrature and detector errors

Cell masses connect a finite source population to an integral on its metric carrier. A covering radius controls nearest-point geometry; an assignment radius controls quadrature for a specified measure. These radii can differ. For example, a golden-orbit point need not lie in its equal-volume permutation cell. The estimates below use the actual cell assignment and its measure.

Let Ω be a metric window with a finite positive Borel measure μ. Choose a finite measurable partition C_i, representatives $s_i \in \Omega$, and masses $v_i = \mu(C_i)$. A measurable convention assigns shared boundaries; null modifications do not affect the statements. Representatives need not belong to their cells. Let f take values in a real Banach space and be integrable on every cell. Set

$$Qf = \sum_i v_i f(s_i), \qquad D_i = \int_{C_i} d(s_i, x)\, d\mu(x).$$

Proposition A.6 (Integrated cell-transport estimate)**.** *Suppose $\|f(s_i) - f(x)\| \le K_i d(s_i, x)$ for almost every $x \in C_i$, where $K_i \ge 0$ and the displacement is integrable. Then*

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \sum_i K_i D_i. \tag{15}$$

If $d(s_i, x) \le H_i$ almost everywhere on C_i, the right-hand side is at most $\sum_i v_i K_i H_i$. In particular, for a globally K-Lipschitz function and a uniform assignment bound H,

$$\left\| Qf - \int_\Omega f\, d\mu \right\| \le \mu(\Omega) K H. \tag{16}$$

For a sequence of such partitions of the same measured window, $H_n \to 0$ therefore gives $Q_n f \to \int_\Omega f\, d\mu$ for each integrable Lipschitz test function. Neither nesting nor a fixed number of representatives is required.

Proof. The difference equals $\sum_i \int_{C_i} [f(s_i) - f(x)]\, d\mu(x)$. The norm of a Bochner integral is bounded by the integral of the norm. Apply the local Lipschitz bound and sum the resulting inequalities. The uniform estimate uses $D_i \le v_i H_i$ and $\sum_i v_i = \mu(\Omega)$. Its right-hand side tends to zero under the stated refinement. The first-moment coefficient is sharp: on $[0, H]$, with Lebesgue measure, one representative at zero and $f(x) = x$, the error and displacement integral both equal $H^2/2$. □

A Lean formalization [8] proves the actual Bochner-integral estimates for finite cell measures, their specialization to measurable disjoint partitions, and convergence under a vanishing assignment bound. It also checks that a constant detector exposes any error in total assigned mass. The source coordinates, partition geometry and their displacement bounds must be established for each population. These hypotheses are not replaced by an assumed quadrature limit.

For the golden permutation cells of side L/q, the established assignment bound $H_q = 2\sqrt{3}L/q$ and equal masses L^3/q^3 instantiate (16). Product time cells of length Δ_q contribute the independent temporal error $TL^3\Delta_q \operatorname{Lip}_t(f)$ on an aligned window of duration T. The spatial term is $TL^3 H_q \operatorname{Lip}_x(f)$. An unmatched endpoint layer contributes at most $L^3\Delta_q \|f\|_\infty$. This application retains the declared event population and time-cell convention; extra routing or bookkeeping events cannot be inserted into its density without another count estimate. The formalization also proves the Fibonacci approximation and floor agreement, constructs the residue-permuted rectangular partition, and derives its equal Lebesgue masses and Euclidean assignment bound. For each fixed globally Lipschitz Banach-valued detector, it proves the actual golden-population quadrature estimate and its refinement limit; the closed and half-open cube integrals agree almost everywhere.

Proposition A.7 (Actual event counts on continuity sets)**.** *Fix $T \ge 0$, $L, c > 0$, $q_n = F_{n+2}$ and $\Delta_n = L/(c\sqrt{q_n})$. Let $C_n(B)$ count the distinct points $(j\Delta_n, s_n(b)) \in B$ with $0 \le j \le \lfloor T/\Delta_n \rfloor$ and $0 \le b_i < q_n$. If B is measurable and its frontier is null for coordinate volume restricted to $\Omega = [0,T) \times [0,L)^3$, then*

$$\Delta_n \frac{L^3}{q_n^3}\, C_n(B) \longrightarrow \operatorname{vol}(B \cap \Omega).$$

The discrepancy between equal event weights and the corresponding clipped time-cell weights is at most $L^3\Delta_n$, including an aligned final endpoint layer whose clipped cell is empty.

Proof. On each cell, replace the indicator of B by its value at the assigned event. Away from the null frontier these indicators converge pointwise, because the assignment distances vanish. They are bounded by one, so dominated convergence gives the clipped-cell count limit. The sum of the nonnegative equal-weight excesses is at most $L^3\Delta_n$; this also bounds the excess over any selected cells. The golden and time-coordinate maps are injective, so the label sums are point counts. □

These steps, the actual Δ_n specialization and the endpoint control are kernel-checked in Lean. Causal frontiers have zero volume by a Fubini argument. The actual count measure converges on moving generated-order intervals; the finite window is part of the reference volume. Strict-pair counts converge to the geometric product measure with the pair endpoint error retained. These proofs are also formalized.

The Lean formalization evaluates the geometric pair measure by Fubini and radial integration. For a timelike diamond contained in the supplied window, the ordering fraction tends to $1/10$; this normalization is formalized as well, with the explicit vertical Alexandrov error, the general

weighted-family limit and the golden specialization. Extra routing events require another count estimate. These theorems select no physical measure or clock.

Proposition A.8 (Localized radial-kernel quadrature)**.** *For $\epsilon > 0$, write*

$$w_x(y) = (1 - d(x,y)/\epsilon)_+, \qquad g_x(y) = w_x(y)[f(x) - f(y)].$$

If f is a real K-Lipschitz function, then g_x is K-Lipschitz and vanishes outside the closed radius-ϵ ball about x. With an assignment map $P(y) = s_i$ on C_i satisfying $d(P(y), y) \le H$, where $H \ge 0$, its quadrature error obeys

$$\left| \sum_i v_i g_x(s_i) - \int_\Omega g_x(y)\, d\mu(y) \right| \le K \int_{\Omega \cap B(x,\epsilon+H)} d(P(y), y)\, d\mu(y) \le KH\, \mu(\Omega \cap B(x, \epsilon + H)). \quad (17)$$

The same statement holds with closed balls; their boundaries are irrelevant for Euclidean volume.

Proof. The tent weight lies in $[0,1]$ and satisfies $|w_x(y) - w_x(z)| \le d(y,z)/\epsilon$. Order the two weights so that $w_x(y) \le w_x(z)$. If both vanish the claim is immediate; otherwise $d(x,z) < \epsilon$. The identity

$$g_x(y) - g_x(z) = w_x(y)[f(z) - f(y)] + [w_x(y) - w_x(z)][f(x) - f(z)]$$

is bounded by $Kd(y,z)[w_x(y) + d(x,z)/\epsilon]$. The bracket is at most one because $w_x(y) \le w_x(z) = 1 - d(x,z)/\epsilon$. The same proof with y, z exchanged covers the other order. If $d(x,y) \ge \epsilon + H$, both kernel terms $g_x(y)$ and $g_x(P(y))$ vanish. Integrate the pointwise bound over the remaining region. The tent Lipschitz estimates have Lean proofs; the displayed support-restricted integral applies the cell-transport argument. □

For three-dimensional volume, multiplying (17) by $45/(\pi\epsilon^5)$ gives the quadrature contribution $60KH(\epsilon + H)^3/\epsilon^5$ in the scalar operator estimate, and hence $480KH/\epsilon^2$ when $H \le \epsilon$. Actual local cell masses or integrated displacements can give smaller bounds than the enclosing-ball volume. They must be computed or bounded for the same assignment. A small fill radius does not justify replacing the equal-volume assignment radius. This estimate alone yields no vanishing operator error when $H = O(q^{-1})$ and $\epsilon = O(q^{-1/2})$.

Dual cells for a local tensor action. A different supplied action can use sorted source coordinates $0 = x_0 < \cdots < x_N = L$, with interval lengths $\ell_j = x_{j+1} - x_j$. Half-interval endpoint weights and interior weights $(\ell_{j-1} + \ell_j)/2$ are the masses of the one-dimensional dual cells. Their tensor products satisfy the preceding transport theorem. For $f \in C^2([0, L])$, linear interpolation on each interval also gives the sharper analytic trapezoidal bound

$$\left| Qf - \int_0^L f(x)\, dx \right| \le \frac{\|f''\|_\infty}{12} \sum_j \ell_j^3 \le \frac{L\ell_{\max}^2}{12} \|f''\|_\infty.$$

Indeed the pointwise interpolation error is at most $\|f''\|_\infty (x - x_j)(x_{j+1} - x)/2$; integrate it on each interval. Successive application in three coordinates gives the tensor bound $L^3 \ell_{\max}^2 \sum_{a=1}^3 \|\partial_a^2 f\|_\infty / 12$. This applies to detector products and sampled mode Gram entries with their stated derivative bounds. Dual-cell weights and equal-volume source-count weights are different finite measures, each with its own comparison to volume. The trapezoidal argument does not improve the radial action by changing its weights implicitly, and it supplies no source-selection law, primitive timing rule or physical calibration.

A.3 Operational selection of the causal cone

The response-selected source Gram space E is three-dimensional. Its unit directions form a sphere, and adjoining a scalar line gives the quadratic form $Q(t,x) = t^2 - \|x\|^2$ on $\mathbb{R} \oplus E$. The finite proper carrier rotations act isometrically on E. Under natural feasibility and objective data with a unique normalized minimizer, the sixty directed seams have equal weight and select the corresponding translation average on the conservative record carrier. These are spatial and algebraic results. The following analytic theorem states an additional condition that selects an operational causal cone on that same carrier.

A displacement cone C contains zero and is closed under addition and nonnegative scaling. Suppose it represents a homogeneous operational order by $p \preceq q$ if and only if $q-p \in C$. Pointedness, $C \cap (-C) = \{0\}$, expresses antisymmetry. Operational covariance means that transformations of admissible histories, interventions and readouts preserve possible influence, with the corresponding linear action on event displacements. Merely changing a frame or relabeling celestial directions does not establish this condition. If C is instead the closed cone generated by finite operations, the conclusion concerns that closure; it does not assert that every displacement is a finite direct read.

For $u \in E$, define

$$X_u(t,x) = (\langle u, x\rangle, tu).$$

For a unit vector n, $B_n(s) = \exp(sX_n)$ is the boost of rapidity s: writing $x = x_\parallel n + x_\perp$,

$$t' = \cosh(s)t + \sinh(s)x_\parallel, \qquad x'_\parallel = \sinh(s)t + \cosh(s)x_\parallel, \qquad x'_\perp = x_\perp.$$

Theorem A.9 (Finite rotations and one boost direction)**.** *Let $G \subset \mathrm{SO}(E)$ be finite and irreducible, as for the proper icosahedral representation. Let $C \neq \{0\}$ be a closed convex pointed displacement cone invariant under* $\mathrm{diag}(1,R)$ *for $R \in G$. Suppose, for one unit n, that $B_n(s)C = C$ for every real s. Then*

$$C = \{(t,x) : t \geq \|x\|\} \quad \textit{or} \quad C = \{(t,x) : t \leq -\|x\|\}.$$

A positive time-axis vector in C selects the first alternative. It is enough to assume boost invariance for a sequence of nonzero rapidities tending to zero.

Proof. Irreducibility gives $\mathrm{span}\{Rn : R \in G\} = E$. Conjugation by $\mathrm{diag}(1,R)$ gives invariance under every $\exp(sX_{Rn})$. Write $u = \sum_j a_j R_j n$. Since $u \mapsto X_u$ is linear,

$$\exp(sX_u) = \lim_{m\to\infty} \left(\prod_j \exp\!\left(\frac{sa_j}{m} X_{R_j n} \right) \right)^m . \tag{18}$$

Indeed, the one-step product and $\exp(sX_u/m)$ have equal constant and linear Taylor terms, so their difference in matrix norm is $O(m^{-2})$. A telescoping sum over m factors bounds the total difference by $O(m^{-1})$; all intermediate products have a uniform exponential norm bound. Every finite product preserves C, and closedness passes this inclusion to the limit. Repeating with $-s$ gives equality. Thus every boost preserves C.

The finite average $|G|^{-1} \sum_{R\in G} R$ projects onto the fixed vectors of G, which are zero by irreducibility. Convexity therefore gives $(t,0) \in C$ whenever $(t,x) \in C$. If $\|x\| > |t|$, boosts parallel to x can make the time coordinate positive or negative. Averaging these two images puts both signs of the time axis in C, contradicting pointedness. Hence every nonzero vector in C is causal.

A nonzero future causal vector has positive time coordinate. Its finite rotation average supplies a positive time-axis vector. Boosts and positive scaling generate every future timelike vector; closedness adds the null boundary. A past vector would similarly supply the opposite time axis,

contradicting pointedness. This gives the future cone. Starting with a past vector gives the past cone. Nontriviality ensures one case occurs.

Finally, if $s_k \neq 0$ tends to zero, for every $s \in \mathbb{R}$ choose integers m_k with $m_k s_k \to s$. Integer powers and inverses of $B_n(s_k)$ preserve C. Continuity and closedness give invariance under $B_n(s)$, reducing the last assertion to the preceding proof. □

The theorem uses only two finite representation facts: the boost-axis orbit spans E, and the group average on E vanishes. It does not infer continuous spatial covariance from the finite carrier action. Its conclusion is a restriction: rotation covariance alone permits every cone $\{(t, x) : t \geq 0,\ \|x\| \leq vt\}$ with $v > 0$, and the stationary time-axis cone. Operational boost covariance excludes those alternatives except the unit-speed cone in the specified scalar coordinate.

The scalar coordinate requires an additive clock readout on actual histories. Once that attachment and the normalized boost action are fixed, the cone slope is fixed relative to them. Constructing the clock attachment is a separate dynamical requirement. The theorem does not select a layer tick, read weights, signal amplitudes or a field equation. The count construction in Section A.4 supplies a separate geometric duration readout on the declared order/count family.

At finite resolution, a bounded event set cannot carry a nontrivial continuous boost action by permutations. A physical use of the theorem therefore needs an operational continuum description or controlled comparisons across a varying family. Finite energy cutoffs and bounded windows must be transformed or restricted consistently. The existing algebraic Lorentz and celestial maps alone supply none of these intervention or clock attachments. The cone classification and product limit above are analytic proofs, rather than Lean formalizations of operational covariance.

A.4 A clock read from source interval counts

The equal-mass source populations of Proposition A.4 admit a clock defined by their records. Suppose an observer can read the complete ancestry of an upper event and identify a lower anchor. Its inclusive interval is $I(p, q) = \{e : p \preceq e \preceq q\}$. Choose a nondegenerate reference interval J in the same history and define

$$\mathcal{T}(I; J) = \left(\frac{|I|}{|J|}\right)^{1/4}. \tag{19}$$

This readout uses order and cardinality. It needs no numerical event labels, spatial coordinates, timestamps, absolute density or oscillator frequency. Complete ancestry access is a substantive readout hypothesis: a hash of an unavailable record does not provide its contents. The cost of collecting and counting the records is separate from the duration reconstructed by the readout.

Theorem A.10 (Count clocks and proper-time ratios)**.** *Consider the declared source histories with their controlled order and equal-mass count limit. Let I_n, J_n approximate fixed interior timelike diamonds, with one common event weight $w_n > 0$, and let their limiting proper-time separations be $\tau_I, \tau_J > 0$. Then*

$$\mathcal{T}(I_n; J_n) \longrightarrow \frac{\tau_I}{\tau_J}.$$

The reference interval fixes the time unit; the clock origin is free.

Proof. For coordinate measure $dt\, d^3x$ and metric $c^2dt^2 - |dx|^2$, a flat diamond of proper-time separation τ has volume $K\tau^4$, where $K = \pi c^3/24$. The vertical formula follows by integrating its ball sections. A Lorentz transformation has determinant one and sends a timelike tip displacement to its rest form, so the same formula holds for moving tips. The count-volume theorem gives $w_n|I_n| \to K\tau_I^4$ and $w_n|J_n| \to K\tau_J^4$. Taking their ratio and its positive fourth root proves the claim. Both the event weight and the volume constant cancel. □

The supplied relation $\Delta = a/c$ enters the proof of the order and volume limit. It is absent from the clock's definition. This distinction allows a retrospective clock to be read from the same history without assuming an oscillator Hamiltonian. It does not select the population, read law or laboratory unit from accepted microscopic repairs.

Proposition A.11 (Finite clock enclosure)**.** *Suppose $A = wN$, $B = wM$ and independently justified volume bounds obey $|A - V_I| \le E_I$, $|B - V_J| \le E_J$, with $B > E_J \ge 0$, $E_I \ge 0$ and $V_J > 0$. Then*

$$\left(\frac{(A - E_I)_+}{B + E_J}\right)^{1/4} \le \frac{\tau_I}{\tau_J} \le \left(\frac{A + E_I}{B - E_J}\right)^{1/4}. \tag{20}$$

The same formula uses counts N, M and errors $E_I/w, E_J/w$.

Proof. The true volumes belong to their stated intervals. Division by a positive denominator and the increasing fourth root preserve the two inequalities. The error bound in (11) supplies such volume errors when its geometric hypotheses hold. Counts alone do not supply those errors. □

For relative volume errors $A = V_I(1 + d_I)$ and $B = V_J(1 + d_J)$, with $|d_I| \le \eta_I < 1$ and $|d_J| \le \eta_J < 1$, the measured clock divided by the exact proper-time ratio lies in

$$\left[\left(\frac{1 - \eta_I}{1 + \eta_J}\right)^{1/4}, \left(\frac{1 + \eta_I}{1 - \eta_J}\right)^{1/4}\right]. \tag{21}$$

Proposition A.12 (Unit freedom within the volume clock class)**.** *A positive elapsed-time function $F(V)$ depending only on diamond volume and additive for successive collinear inertial intervals is $F(V) = \kappa V^{1/4}$ for some $\kappa > 0$.*

Proof. Set $g(t) = F(Kt^4)$ for $t > 0$. Inertial additivity gives $g(t + s) = g(t) + g(s)$. Positivity implies strict monotonicity. Additivity first fixes g on positive rationals; monotonicity and rational approximation give $g(t) = g(1)t$ for all positive real t. This is uniqueness within the stated class, not a premise about every physical pointer. □

Accumulating duration along a curve. Let $\gamma(t) = (t, x(t))$ be C^2 on $[0, T]$, with $|x'(t)| \le v < c$, $|x''(t)| \le A_0$, and a common spatial buffer for its small diamonds. Segment durations are comparable to $\delta_n \to 0$. Suppose

$$\frac{\Delta_n}{\delta_n} \to 0, \qquad \frac{H_n}{\delta_n} \to 0, \qquad \frac{h_n}{a_n} \to 0,$$

and sampled tips have spatial and temporal errors $O(H_n + \Delta_n)$. Summing (19) over successive segment intervals, using the same fixed reference interval, converges to

$$\frac{1}{\tau_J} \int_0^T \sqrt{1 - |x'(t)|^2/c^2}\, dt. \tag{22}$$

Indeed, the two-shell estimate is uniformly $O(\delta_n^4 h_n/a_n + \delta_n^3(H_n + \Delta_n))$ per segment. The strict speed margin bounds each exact diamond volume below by a positive constant times δ_n^4. Tip perturbations have the same relative error order. Equation (21) therefore controls all segment clocks by one vanishing relative error.

The continuum polygon error has an explicit bound. For $f(u) = \sqrt{1 - |u|^2/c^2}$ on $|u| \le v$, its Hessian norm is at most $M_0 = [c^2(1 - v^2/c^2)^{3/2}]^{-1}$. Taylor expansion about each segment's average velocity has zero integrated linear term. Since the velocity differs from that average by at most $A_0\delta_i$, the chord and curve proper times differ by at most $M_0 A_0^2 \delta_i^3/2$. The total is at most $M_0 A_0^2 T(\max_i \delta_i)^2/2$. This proves (22). Fixed one-layer intervals do not satisfy these resolution hypotheses.

Finite records and calibration scope. The finite executable control parses writer identities, immutable register versions and values in the $q = 5$ history. Its 500 events give the three centered inclusive counts $2, 41, 80$. The first two continuum diamonds lie inside the source window; the third does not. Exact dyadic fourth-root bounds, schedule permutation and opaque event relabeling check the readout. These coarse intervals provide no finite proper-time accuracy certificate. The source-pinned expected hash authenticates the replayed content relative to that commitment; it is not an external signature or a proof of physical provenance. Unrelated serial audit links are excluded from the semantic read order.

Lean formalizes the finite interval and root inequalities. The count-volume and curve limits above are analytic proofs. In this layered law every causal pair k layers apart has a path of exactly k reads, and every strict chain has at most k edges. Longest-chain height consequently reads coordinate duration independently of displacement. It is not a proper-time clock for moving tips in this family. The count construction concerns geometric duration reconstructed from accessible records; it neither sets the time cost of primitive seam words nor identifies physical clock readings without a common physical interpretation of the event law.

A.5 Source selection and a sufficient record-transport law

The source-selection question has two different meanings. A unique finite golden population and radius are stronger requirements than a common operational causal/count limit. Here we prove a sufficient source law for the latter, and compare M1 as one regulator of it. The proposed law is *additional* to A1–A3. It is neither adopted as a fourth core axiom nor asserted to hold in the registered simulator. Its several substantive requirements are stated together; no claim of minimality or unique necessity is made.

Why the existing data-access clauses do not supply a channel. There is a scalar-seam construction with a complete port response, rather than a response-free scalar toy. At each vertex of the normalized midpoint sphere S_n put

$$\mathcal{A}_i = \bigoplus_{p=1}^{12} M_6(\mathbb{C}), \qquad \mathcal{R}_i = Z(\mathcal{A}_i) = \mathbb{C}^{12}.$$

Its twelve block units are primitive central ports. Distinct-carrier seams are $\mathbb{C}$, with scalar inclusion and the unital normalized trace $E_i(a) = \sum_p \mathrm{Tr}(a_p)/72$ as visible data restriction. Triple restrictions are identities. Local chart overlaps are distinguished from these inter-carrier seams. Their proper action groupoid implements all sixty carrier rechartings; it does not transmit a private record to another carrier.

An exact response on the same accessible internal $\mathbb{C}^6$ is

$$D_p = \mathrm{diag}\Big(\widehat{v_p/2} + iQ_p, \widehat{\sigma(v_p)/2}\Big), \qquad Q_p = \frac{v_p v_p^{\mathsf{T}}}{2} - \frac{r^2}{6} I + \frac{1}{12} I,$$

where v_p is an icosahedral calculation frame, $r^2 = \|v_p\|^2$, σ is its Galois conjugation and $\widehat{v}w = v \times w$. Even and odd antipodal fields span respectively the six imaginary symmetric directions and the two three-dimensional real skew spaces. Thus the response is injective, commutator closed and has image $\mathfrak{u}(3) \oplus \mathfrak{so}(3)$. Full M_6 effects observe its relative block phase; restricting to block-diagonal effects loses one observable response direction. For each proper action a, $U_a = \mathrm{diag}(R_a, \sigma(R_a))$ satisfies $U_a D_p U_a^* = D_{ap}$. Executed Cayley factors in the original response span realize these

matrices, their covariance and their compositions. The central permutation is the chart map; an inner automorphism of a carrier algebra is not claimed to permute its center.

The admitted instruments are local: central readout, destructive internal effect readout into two central ports, requested-block trace reset, prediction from a supplied preparation and local conditioning, and in-place checkpoint continuation. Only the declared response span generates infinitesimal reversible port operations. Irreversible read/reset channels have kernels and add no reversible direction. Independent ticks carry no record-dependent controller information between carriers.

The open-star nerve is S_n, the bridge is the identity and its fundamental cycle has degree one. Midpoints coarsen to declared endpoints, transported under relabeling; fundamental chains push forward exactly and spherical mesh contracts. Coarse local states average their descendant fibers. The instrument lift runs independently on every descendant, with each branch's own outcomes, then uses one fixed local tag through continuation. Selective conditioning updates tag weights; longer refinements use the composed weights. The tagged direct-sum experiment is not misidentified with the descendants' joint tensor algebra or with a communication channel.

The feasible local-state family is unconstrained except for normalization and chart covariance. Its complete grammar contains all 431 independent Hermitian state coordinates per carrier. The declared reference is $I_{72}/72$, the cover contains every carrier, and the weight is $1/N_n$. The weighted Umegaki objective has the unique tracial minimizer by nonnegativity and its equality condition. Every omitted coordinate admits two positive trace-one perturbations invisible to the remaining coordinates. This is an explicit A3 problem, not invariance used to choose a reference among all possible source ontologies.

Induction on every admitted local operation shows that changing an independent private source port leaves another carrier's entire local transcript unchanged. The induction permits arbitrary receiver-local memory and stopping rules, so no decoder of that transcript reads both interventions correctly. For a uniform unknown port its success probability is at most $1/12$. Hence the stated typed data-access clauses, with local chart holonomy distinct from inter-carrier record transport, do not entail M1's remote read service. The detailed full-clause interpretation and all-level refinement proof are in `code/source_selection_model/DERIVATION.md`; finite receipts do not replace that analytic model-membership argument. If reversible overlap holonomy is strengthened to require inter-carrier record transport, this particular countermodel is excluded by that additional requirement.

Proposed law RG: response-natural classical record transport. The complete-response theorem for A1–A2 gives $\mathfrak{su}(3) \oplus \mathfrak{su}(2) \oplus \mathfrak{u}(1)$ at a complete reference carrier. Its unique three-dimensional simple ideal E, with the response trace form, has connected adjoint action $SO(E)$. This is an internal Euclidean response space. RG says that the central-record source admits a complete, intervention-faithful, refinement-compatible realization of the following generated process theory, graded by a source-observable duration:

1. Objects are arbitrary finite independent classical registers at addresses in the affine space modeled on E. Finite tensor products, local preparation, finite deterministic classical maps, nondestructive read/copy, discard and retained versions are realizable at finite cutoff. Their ideal displacement and local duration are zero. Their actual implementations and storage costs are retained.

2. One retained-record flight has displacement e, $\|e\| = 1$, and duration $\tau > 0$. Translation, connected response rotation and positive scaling generate actual flights $(\lambda Re, \lambda\tau)$ for $R \in$

$SO(E)$ and $\lambda > 0$. Waiting preserves the record. These processes transmit retained records between carriers.

3. Sequential and finite parallel composition are realized. Time adds on dependency paths; independent branches share elapsed time. Every native dependency, including controller, timing and metadata channels, factors through the graded local/flight/wait grammar. Conversely every finite generated process has an admitted native realization preserving all independent record interventions and immutable versions, with every intermediate event and its resource bill retained. Realizations commute with the declared source refinements.

Intervention faithfulness includes local interventions at every designated intermediate record output before publication, with all other registers and cached copies held fixed. Lowering must commute with insertion of these operations and subsequent reads. The version is committed once after its intervention. Merely agreeing on final outputs for all initial preparations is weaker: two versions can both contain $x + 1$, making a substituted read indistinguishable until one version alone is perturbed.

The exact version realizes these ideal grades. The vanishing-overhead version requires, for every complete finite experiment and every $\epsilon > 0$, a finite realization with exact record values and endpoint addresses, no faster native dependence, and each logical event time in $[t, t + \epsilon]$ relative to its ideal schedule. This uniform whole-program requirement is part of RG. Nonnegative primitive overheads summing to at most ϵ on every complete path suffice, by induction on the earliest-time recurrence using addition and maximum. A fixed error at each layer is not a substitute.

RG supplies flat homogeneous operational addresses, record protection, cofinal workspace, parallel composability and a physical connection from the internal response to record motion. None follows just from internal rotation covariance. Translations and scales act on a family of finite experiments, possibly changing cutoff; they are not extra infinitesimal port generators on one carrier. A proposed implementation must obey A1's full response constraint. A3 can select states inside such a source but is not used to derive RG or compel a particular observer program.

Proposition A.13 (RG suffices for the operational M1 limit)**.** *Under the complete-response conclusion of A1–A2 and RG, there exist cofinal finite independent central-record experiments whose native ancestry on their distinguished records has the round* $3 + 1$*-dimensional causal limit, continuity-set volume counts, Alexandrov interval counts and four-dimensional ordering fraction. M1's golden population with* $q = F_n$ *and observation duration* $L/(c\sqrt{q})$ *is one regulator with these limits, where* $c = 1/\tau$*. Its finite radius is not a uniquely selected microscopic length.*

Proof. Each primitive flight has norm c times its duration. The triangle inequality bounds every complete native dependency by $\|y - x\| \le c(s - t)$, including paths through auxiliary events. Conversely, for $\|v\| \le cT$, response transitivity rotates e to $v/\|v\|$. Scaling by $\|v\|$ and waiting for $T - \tau\|v\|$ realizes the read. The zero displacement uses waiting. The exact process has precisely the closed round cone; vanishing overhead attains every strict timelike read and excludes every spacelike read. Finite tensor composition realizes any finite list simultaneously, with protected source versions.

In a calculation frame prepare records at $x_b = Lb/q$, $b \in \{0, \dots, q-1\}^3$. RG supplies their joint finite preparation, rather than a presumed native carrier census. Half-open cells have equal volume $(L/q)^3$ and assignment/covering error $h_q \le \sqrt{3}L/q$. Choose $\delta_q \to 0$ with $q\delta_q \to \infty$. Exhaustively observe the preceding population within duration δ_q, then commit the consumed source versions. The primitive latency calculation derives the complete read menu $\|x_b - x_a\| \le c\delta_q$; this menu is not input to the source law. The covering-path theorem constructs a k-layer read path whenever

$$\|y - x\| \le k(c\delta_q - 2h_q), \qquad \text{and every generated path obeys} \quad \|y - x\| \le kc\delta_q.$$

Independently variable records and retained versions make these actual causal dependencies. Compiling a primitive correctly on every input state extends to every finite word by induction; concatenating the native words retains the sum of their event bills.

For each bounded horizon, lower the entire finite experiment with uniform time error $\epsilon_q \le \delta_q/q$. Commits need not be perfectly simultaneous. Let P_q be the consumed-writer order and N_q full native ancestry restricted to those commits. For each consumed-writer edge $u \to v$, vary the record at u's output boundary while holding other records fixed. The corresponding tuple coordinate read at v changes. If no native path connected them, the ancestors of v would form a dependency-closed region unchanged by that local intervention. Induction on the native updates would leave its read unchanged, a contradiction. This proves $P_q \subseteq N_q$ using intermediate intervention faithfulness; the value-only compilation induction does not prove it. Complete native grading then gives

$$P_q \;\subseteq\; N_q \;\subseteq\; C_c \quad \text{at the actual native event assignments.}$$

Every fixed strict timelike comparison eventually belongs to P_q, while every fixed spacelike or backwards comparison is eventually excluded from N_q. Moving two positions by at most H and times by at most ϵ changes the causal margin by at most $2H + 2c\epsilon$; this tends to zero.

Space and time cells give distinguished-event weight $w_q = (L/q)^3\delta_q$. For any bounded continuity set A in the window, its pulled-back indicator converges almost everywhere and is bounded by one, hence $w_q \# A_q \to \int_A dt\, d^3x$. A clipped terminal layer has mass error at most $L^3\delta_q$. For intervals, use the order sandwich and the null measure of the endpoint cone boundaries. For strict pairs, apply the same argument on the product window to both interval indicators and the precedence indicator; the time diagonal is also null. Dominated convergence gives the limits for both P_q and N_q, without excluding additional native edges by assumption. An interior rest diamond of duration T has $V = \pi c^3 T^4/24$ and oriented-pair measure $V^2/20$. Consequently $2\,\#\text{strictPairs}/[N(N-1)] \to 1/10$.

For the golden regulator use the proved equal-volume assignment with $h_q \le 2\sqrt{3}L/q$ and $\delta_q = L/(c\sqrt{q})$. RG supplies the finite preparations and transmissions, and all previous bounds apply. The radius $L/\sqrt{q} = c\delta_q$ is the reach of the chosen observation clock. Both regulators thus converge to the same causal/count observables. For a K-Lipschitz observable, assignment errors (H, ϵ) change its weighted integral by at most $K(H + \epsilon)$ times the window volume, plus the clipped endpoint term. Comparing regulators adds their bounds; indicator comparisons use null boundary tubes instead. □

This conclusion concerns a realizable observer probe and its operational limit. It neither selects spontaneous execution of that probe nor equates its distinguished-event volume with a count of all microscopic repair events. It proves no Lorentz covariance of matter, curved geometry or laboratory clock calibration. Finite graphs need not be isomorphic, and unrelated field dynamics do not inherit equivalence from count convergence.

Finite clearance, verification and deletion controls. The golden squared displacements in units of L^2 lie in $\mathbb{Z}[\phi]$. For integer $q > 1$ none equals $1/q$: irrationality would first force the coefficient of ϕ to vanish, leaving an integer equal to $1/q$. Thus every admitted nonself golden flight has positive finite slack, although controller/commit costs remain charged. The rational-grid reference uses $L = c = 1$ and $\delta_q^2 = 2/(2q+1)$; boundary equality would require $(2q+1)m = 2q^2$, impossible since $\gcd(2q+1, q) = 1$.

The reference prepares, forks, flies, waits, adds locally and commits two layers for $q = 2, 3, 5$. Baselines, every single-site initial intervention and every first-layer commit intervention retain 323 executions, totaling 5,344,106 primitive events. Compact commitments are checked against live

streams and independent semantic replay, including all consumed writers and intervention path multiplicities. These are executions of RG's reference grammar, not of an asserted native simulator implementation. Source pins and hostile-input tests reject forged timing, values, missing events and filtered intervention denominators. The additive reference diagnostic is distinct from the proof's lossless tuple payload, constructed analytically from RG's finite classical maps. Its intervention tests certify that each consumed value affects the diagnostic; they do not certify arbitrary payload maps or native programs.

The Lean proofs cover the local information bound, the primitive cone and its covering paths, the lowering induction, the localized-intervention ancestry reduction, event bills, population size and margin stability. The full model interpretation and the join from RG to native ancestry and counts are analytic. Detailed proofs appear in `code/source_selection_model/RECORD_GLUING.md`.

Deleting transport gives the scalar-seam control; keeping only axis flights gives an ℓ^1 cone; an ungraded metadata channel defeats the upper cone bound; destructive transfer or absent parallel workspace defeats the finite experiment. These controls explain the content of the proposed law instead of concealing it in a successful M1 schedule.

B Fields, volume and entropy on source populations

B.1 Spatial populations and a common scalar continuum action

The completed conservative source-record carrier is a Euclidean three-space under the stated response and metric hypotheses. Its whole dense record group is not a locally finite population. A finite-resolution selection and a transport law can be stated on that carrier without inserting a Cartesian lattice. This construction keeps both choices explicit.

Fix $r, \delta, \eta > 0$, a response-metric ball $\Omega = \overline{B}(0, r)$, and a finite directed seam-word endpoint menu $M \subset \Omega$ that is an η-net of Ω. Such menus exist: density approximates a slightly inward-shifted point by a source endpoint, compactness extracts finitely many approximants, and a finite maximum word budget contains all of them. This is algebraic endpoint availability, not a proof that a repair law accepts those words as moves. Choose a maximal δ-separated subset $S \subset M$, and set $h = \delta + \eta$. The resolution, window and tie rule are supplied; maximal means inclusion-maximal, not maximum cardinality or a selected energy minimum.

Proposition B.1 (Finite spatial population and metric transport)**.** *The selected points form an h-net, and*

$$(r/h)^3 \le |S| \le (1 + 2r/\delta)^3.$$

For $R \ge h$ and $B(x, R) \subset \Omega$,

$$((R - h)/h)^3 \le |S \cap B(x, R)| \le (1 + 2R/\delta)^3.$$

Connect distinct points whose source distance is at most $a > 2h$, weighting edges by that distance. The graph is connected, degrees are bounded by $(1+2a/\delta)^3 - 1$, and its shortest-path metric satisfies

$$|s - t| \le d_G(s, t) \le \left(1 + \frac{2h}{a - 2h}\right) |s - t| + 2h.$$

Consequently $a \to 0$ and $h/a \to 0$ give uniform metric convergence on the fixed window. With $\eta \le \delta$ and $a = 8\delta$, degrees instead stay at most 4912 and $d_G \le 2|s-t| + 4\delta$; this latter estimate alone does not imply a Euclidean metric limit.

Proof. Maximality puts each menu point within δ of S, proving coverage. Disjoint radius-$\delta/2$ balls give the upper counts; covering the window, or $B(x, R-h)$, by radius-h balls gives the lower counts. For the graph, divide the segment from s to t into $k = \lceil |s-t|/(a-2h) \rceil$ pieces. Convexity retains the segment in Ω; approximate its interior division points within h, keeping its endpoints exact. Consecutive approximants are at most a apart and the path length is at most $|s-t| + 2hk$. The triangle inequality gives the lower bound. These estimates are independent of the choice among maximal sets. □

The metric fixes relative three-volume. Let C_i be the Voronoi cell of $x_i \in S$, clipped to Ω, and $v_i = \mathrm{vol}(C_i) > 0$. Thus $\sum_i v_i = V = \mathrm{vol}(\Omega)$, and almost every point of C_i is at distance at most h from x_i. These are metric weights, not a physical event-count density. For $0 < h \le \epsilon$ supply

$$a_{ij} = \frac{45 v_i v_j}{\pi \epsilon^5} (1 - |x_i - x_j|/\epsilon)_+, \qquad (Lq)_i = v_i^{-1} \sum_j a_{ij}(q_i - q_j).$$

The weighted inner product is $\langle p, q \rangle_v = \sum_i v_i p_i q_i$. Symmetry gives $\langle q, Lq \rangle_v = \frac{1}{2} \sum_{ij} a_{ij}(q_i - q_j)^2 \ge 0$. The common finite action is

$$S_h[q] = \frac{1}{2} \int \left[\|\dot{q}\|_v^2 - c^2 \langle q, Lq \rangle_v - m^2 \|q\|_v^2 \right] dt, \qquad c > 0, \quad m \ge 0. \tag{23}$$

Its equation is $\ddot{q} + c^2 Lq + m^2 q = 0$. The coefficient m has inverse-time units in this convention; neither physical mass nor time units are inferred.

Proposition B.2 (Uniform scalar and detector limit)**.** *Let $u_{tt} - c^2 \Delta u + m^2 u = 0$ be a smooth reference on the full carrier. On $[0, T]$, its spatial support has distance greater than ϵ from the complement of Ω. Write $M_1 = \sup |\nabla u|$ and $M_4 = \sup \|D_x^4 u\|_{\mathrm{op}}$, with suprema over the full carrier and time window. Then*

$$\left| (Lu|_S)_i + \Delta u(x_i) \right| \le \frac{15}{112} \epsilon^2 M_4 + \frac{120 h (\epsilon + h)^3}{\epsilon^5} M_1 \le \frac{15}{112} \epsilon^2 M_4 + \frac{960 h}{\epsilon^2} M_1.$$

Put $B = c^2 \sqrt{V} [(15/112)\epsilon^2 M_4 + 960 h \epsilon^{-2} M_1]$. Initialize the finite action with the sampled reference position and velocity. Then

$$\|\dot{q} - u_t|_S\|_v \le tB, \qquad \|q - u|_S\|_v \le \tfrac{1}{2} t^2 B.$$

For a fixed C^1 detector b, the field readout $D_h = \sum_i v_i b(x_i) q_i$ and its continuum counterpart $D = \int_\Omega bu$ obey

$$|D_h(t) - D(t)| \le \tfrac{1}{2} \sqrt{V} \|b\|_\infty t^2 B + V h \|\nabla(bu(t))\|_\infty.$$

Thus $\epsilon \to 0$ and $h/\epsilon^2 \to 0$ yield a uniform finite-time observable limit. In normalized carrier units $\epsilon = h^{1/4}$ gives an $O(\sqrt{h})$ bound. Nested populations are unnecessary.

Proof. Compare L with the integral operator having kernel $45(1 - |z|/\epsilon)_+/(\pi\epsilon^5)$. Odd Taylor terms cancel and

$$\int (1 - |z|/\epsilon)_+ z_1^2 \, dz = \frac{2\pi}{45} \epsilon^5, \qquad \int (1 - |z|/\epsilon)_+ |z|^4 \, dz = \frac{\pi}{14} \epsilon^7.$$

The second moment gives $-\Delta$; the fourth-order Taylor remainder, bounded by $M_4 |z|^4/24$, gives $15\epsilon^2 M_4/112$. For nearest-cell projection P, compare $g_x(y) = (1 - |x-y|/\epsilon)_+ (u(x) - u(y))$ with $g_x(P(y))$. Splitting the difference at the larger kernel value uses a point within ϵ of x; kernel and field Lipschitz bounds give $|g_x(P(y)) - g_x(y)| \le 2hM_1$. This difference vanishes outside $B(x, \epsilon + h)$,

giving the stated quadrature term. If $u(x) \neq 0$, the support buffer retains the whole kernel ball in Ω; otherwise the integrand outside Ω is zero. No boundary term is omitted.

The sampled residual in the weighted norm is at most B. The positive operator $A = c^2L + m^2I$ has $\|\cos(t\sqrt{A})\| \leq 1$ and $\|\sin(t\sqrt{A})/\sqrt{A}\| \leq t$, with the zero eigenspace interpreted by continuity. Duhamel integration gives the two trajectory bounds. Cauchy–Schwarz and nearest-cell quadrature for bu give the detector bound. □

The same cell argument gives $\|L\|_v \leq 960/\epsilon^2$, using twice the largest weighted degree. `MetricKernelEnergy` formalizes the finite positive-energy and mass-weighted degree inequalities; the geometric, integral and continuum arguments above are analytic proofs. A local discrete-action time integrator with a strict Courant bound may consume this operator and add its own time and initialization errors.

Smooth compact initial data admit the reference until their speed-c domain of influence reaches the window boundary. Local energy flux for the Klein–Gordon equation gives that speed; the nonnegative mass term changes energy density but not the characteristic cone. Applied to a smooth local preparation difference, the detector estimate bounds the finite response outside that cone. A nonzero continuum signal is resolvable when its lower bound exceeds the approximation and detector errors. Instantaneous small tails of the finite graph are not identified with detectable superluminal signals. This is an interior effective limit, not a theorem about arbitrary boundary conditions or exact microscopic order reflection.

The canonical variables $Q_i = \sqrt{v_i}q_i$ and $P_i = p_i/\sqrt{v_i}$, $p_i = v_i\dot{q}_i$, give the oscillator Hamiltonian $\frac{1}{2}(|P|^2 + Q^T\widetilde{A}Q)$, where $\widetilde{A} = M^{1/2}(c^2L + m^2I)M^{-1/2} \geq 0$ and $M = \mathrm{diag}(v_i)$. Supplied canonical quantization on $L^2(\mathbb{R}^{|S|}, dQ)$ gives the same field readouts as multiplication operators. For $m > 0$ every mode has a Gaussian vacuum; at $m = 0$ zero modes remain free particles unless a separate restriction is declared. This finite Hilbert construction is not an interacting quantum continuum or a derivation of the Standard Model.

Coarsening, the volume prescription, radial force law, inertia, action coefficients and accessible observer variables remain added assumptions. The kernel graph may have growing degree. Algebraic source words alone do not implement its couplings as accepted primitive repairs or calibrate their duration. The construction supplies one conditional scalar effective world on the response carrier; its selection and physical interpretation remain separate obligations.

Proposition B.3 (Controlled boost comparison for scalar detectors)**.** *Let Λ be a proper orthochronous Lorentz transformation of $c^2dt^2 - |dx|^2$, and let u be a smooth Klein–Gordon solution on the full spacetime. Put $u^\Lambda = u \circ \Lambda^{-1}$. Suppose both references satisfy the support and derivative hypotheses of Proposition B.2 on the same $[0,T]$ and Ω, with a common positive interior buffer. Choose a real $f \in C^1_c(\mathbb{R}^{1+3})$ such that both f and $f^\Lambda = f \circ \Lambda^{-1}$ are supported in $(0,T) \times \Omega^\circ$. For $w = u, u^\Lambda$, let q_h^w solve the same action (23), with the same L, v_i, c, m, initialized by $w(0, x_i)$ and $\partial_t w(0, x_i)$. Define the spacetime-smeared readouts*

$$\mathcal{D}_h(f; q) = \int_0^T \sum_i v_i f(t, x_i) q_i(t)\, dt, \qquad \mathcal{D}(f; w) = \int_{\mathbb{R}^{1+3}} f w\, dt\, d^3x.$$

Using the preceding spatial derivative bounds $M_1(w), M_4(w)$, put

$$B_w = c^2\sqrt{V}\left[\frac{15}{112}\epsilon^2 M_4(w) + 960h\epsilon^{-2}M_1(w)\right],$$

and

$$\mathcal{E}_h(f, w) = \frac{\sqrt{V}B_w}{2}\int_0^T t^2\|f(t,\cdot)\|_\infty\, dt + Vh\int_0^T \|\nabla_x(f(t,\cdot)w(t,\cdot))\|_\infty\, dt.$$

Then

$$\left|\mathcal{D}_h(f^\Lambda; q_h^{u^\Lambda}) - \mathcal{D}_h(f; q_h^u)\right| \le \mathcal{E}_h(f^\Lambda, u^\Lambda) + \mathcal{E}_h(f, u). \tag{24}$$

The right-hand side tends to zero for $\epsilon \to 0$ and $h/\epsilon^2 \to 0$. The conclusion is uniform over a bounded boost class when the common window, buffer and displayed derivative and detector bounds are uniform over that class.

Proof. Lorentz covariance of the Klein–Gordon equation makes u^Λ a solution with the same c, m. The preceding detector estimate applies at each time with $b = f(t, \cdot)$; its time integral is $|\mathcal{D}_h(f; q_h^w) - \mathcal{D}(f; w)| \le \mathcal{E}_h(f, w)$. Since $\det \Lambda = 1$ and both detector supports lie in the comparison window, change of variables gives $\mathcal{D}(f^\Lambda; u^\Lambda) = \mathcal{D}(f; u)$. The two approximation bounds and the triangle inequality prove the claim. □

The transformed preparations are supplied through the continuum reference on the transformed Cauchy surface; this comparison does not construct a boost map from one finite history or its initial data. The same estimate applies to smooth local preparation differences by linearity. Spacetime smearing is essential to the displayed comparison: a fixed-time detector instead transforms to a tilted hypersurface with its transformed measure. The result controls detector values for one common declared action, not exact finite-graph support, native source selection, or boosts of arbitrary finite records. Time integration and preparation errors, if present in an execution, require their own bounds.

B.2 Action measure, record production and geometric density

A supplied scalar response operator contains more information about its action measure than an unweighted event count. It does not identify that measure with physical volume. The distinction can be tested on the same source-addressed family, with its preparation, protected records, read law and model clock kept explicit.

Proposition B.4 (Identification of normalized action measure)**.** *Let A be a finite real operator whose nonzero off-diagonal support is connected. Suppose a positive diagonal matrix $M = \mathrm{diag}(m_i)$ makes MA symmetric. The equations*

$$m_i A_{ij} = m_j A_{ji}, \qquad \sum_i m_i = 1$$

determine m uniquely. If the two directed coefficients on every support edge are negative, a spanning tree constructs the candidate through $m_j/m_i = A_{ij}/A_{ji}$; the remaining edges test consistency.

Proof. For two positive solutions m, n, multiplying their edge equations and cancelling a nonzero coefficient gives $m_i n_j = m_j n_i$. Thus m_i/n_i is constant along every path. Connectivity makes the constant global, and normalization sets it to one. Tree products produce the candidate; positivity and every remaining edge equation certify its extension to the whole graph. Multiplying both M and the stiffness MA by one positive constant leaves A unchanged, so reciprocity cannot identify an absolute action scale. □

This reconstruction also has a finite conditioning bound. If each directed edge coefficient has relative error at most $\epsilon < 1$, each inferred edge ratio changes by a factor between r^{-1} and r, where $r = (1+\epsilon)/(1-\epsilon)$. For a tree of depth d, the root-relative weights change by factors in $[r^{-d}, r^d]$. Normalizing their sum gives factors in $[r^{-2d}, r^{2d}]$. This bounds the tree estimate; noisy cycle disagreement remains a model-consistency test.

On the authenticated $q = 5$ scalar family, the independent certificate recovers all 64 normalized weights from the directed operator alone, checks its 144 undirected edges, and compares the result with the separately reconstructed golden dual-cell quadrature. The tree has depth 9 and 81 remaining cycle edges. The parent mass vector is used for this comparison, not by the reconstruction algorithm. Operator and action selection remain supplied inputs.

A production control on the same finite family. Two preparations, with amplitude multipliers 1/2 and 1 relative to the retained source, give 2,944 independently replayed field writes. They have the same response operator and field ancestry; their initial energies differ by a factor of four. Local observer records read the actual writer, immutable version and value at the same site. Their extra read/write operations are charged separately and have no identified physical durations. The union of these local certificates is not a construction of remote record transport to one observer.

For an interior field-ancestry interval I, let $v_e = \tau m_{i(e)}$, where $\tau = \sqrt{5}/35$ is the retained action tick. Declare a production multiplier $r > 0$ and an integer exposure budget K. The finite local production rule

$$n_e = \lfloor K r v_e \rfloor, \qquad \widehat{V}_r(I) = \frac{1}{Kr} \sum_{e \in I} n_e$$

has the exact deterministic enclosure

$$0 \le \sum_{e \in I} v_e - \widehat{V}_r(I) < \frac{|I|}{Kr}. \tag{25}$$

The proof sums $0 \le K r v_e - n_e < 1$. Four controls cross the two source strengths with $r = 1, 2$ at $K = 4096$. All 324 local observer records are replayed. Changing r changes raw density while the field order and action measure stay fixed; dividing by the declared production law recovers the same finite measure within (25). Neither that production law nor the identification of τm_i with physical four-volume is derived by this test. No Poisson reference or old causal count-clock theorem is substituted for the actual field ancestry.

What a specified conformal action adds. Consider the separate control metric

$$g = u(x)\,(-dt^2 + dx^2 + dy^2 + dz^2), \qquad u > 0,$$

and the canonical massive scalar action in these coordinates. Its kinetic weight scales as u, whereas its metric four-volume scales as u^2. In the finite control, $M_i(u) = u_i M_i(1)$, each undirected gradient coefficient is multiplied by $(u_i + u_j)/2$, the Dirichlet boundary coefficient by its interior u_i, and the unit-mass potential coefficient by u_i^2. These terms define one symmetric quadratic action and reproduce the original operator at $u = 1$.

Reciprocity determines the relative profile u_i/u_j after division by the reference weights. The full operator supplies more: at an interior site with all six neighbours and the specified unit mass potential,

$$\sum_j A(u)_{ij} = u_i.$$

Gradient terms cancel in this row sum. It calibrates the common profile factor; all eight interior sites provide independent consistency checks. Thus no additional value of u is imported into this reconstruction. The unit scalar mass and the conformal action class are essential inputs. The controls $u = 1$ and $u = 1 + x/2$ recover both profiles exactly while retaining the same nonzero

support and hence the same layered support order. Their normalized kinetic and four-volume measures differ for the nonconstant profile.

For an identified affine control $u = a + bx$, its specified continuum metric has scalar curvature

$$R = -6\Omega^{-3}\Box\Omega = \frac{3b^2}{2u^3}, \qquad \Omega = \sqrt{u},$$

with signature $(- + ++)$ and the displayed curvature convention. The nonzero control curvature is obtained through the supplied action and metric dictionary, not from an unweighted ordering statistic. No law makes either profile respond to the simulated source strength.

Scalar readouts do not determine tensor response. The same finite action also permits a direct comparison of directional responses. Write its potential as $V = G_x + G_y + G_z + V_m$, including every interior and Dirichlet edge, and consider the retained neighbouring configurations q_-, q. Their discrete action contribution is

$$L_d = T_d - \tau V, \qquad T_d = \frac{(q - q_-)^T M (q - q_-)}{2\tau}.$$

The potential uses the right endpoint q. Summing these contributions gives the recorded centred update through the discrete Euler–Lagrange equation. At fixed configurations, a stretch of coordinate axis k by s changes the contribution to

$$L_d(s) = sT_d - \tau\{s(V - G_k) + G_k/s\}, \qquad \left.\frac{dL_d}{d\log s}\right|_{s=1} = T_d - \tau(V - 2G_k).$$

An x/y permutation commutes with the complete cubic source operator. It preserves T_d, V, the action measure and the isotropic sum of these derivatives, but interchanges unequal directional responses. The certificate uses the two retained configurations and independently reassembles the stretched-grid action at $s = 2$ and $s = 1/2$; two thirds of their difference equals the displayed derivative exactly. Kinetic variation is included. These are finite action responses, with no identification of a physical stress source or gravitational coupling.

The connectedness and normalization theorem, common-scale ambiguities, finite production-error bound and conformal calibration algebra are checked in Lean; the source histories, exact controls and an independent verifier accompany it [8]. They identify a relative action measure and a profile within a specified conformal action class. They do not select the population, action, clock, production grammar, storage, source-to-metric update or physical tensor coupling. Those inputs are needed before the recovered quantity can be used as a sourced physical volume or an Einstein response.

B.3 Calibrated clocks and an undetermined light-deflection coefficient

The source action measure and a readable scalar clock do not by themselves select the spatial gravitational response. A constructive comparison uses the grounded massless stiffness $K = M(\mathsf{A} - I) > 0$ on the same prepared golden addresses, with two real fields u, v and an identical source j. The constitutive law, source preparation, update program and physical probe dictionary are declared. They are not consequences of A1–A3.

Proposition B.5 (Two-field clock nonidentifiability)**.** *For every $\kappa > 0$ and $\gamma \in \mathbb{R}$, let*

$$E_{\gamma,\kappa}(u, v; j) = \frac{1}{2}u^{\mathsf{T}}Ku + \frac{\kappa}{2}(v - \gamma u)^{\mathsf{T}}K(v - \gamma u) - j^{\mathsf{T}}u. \tag{26}$$

Every positive symmetric two-field edge matrix whose clock Schur complement equals one has this form. The unique stationary fields satisfy $u = K^{-1}j$ and $v = \gamma u$. The entire clock susceptibility, Gaussian clock marginal at any fixed positive inverse temperature, and clock dynamics under the kinetic form $[\dot{u}^{\mathsf{T}} M \dot{u} + \kappa \dot{w}^{\mathsf{T}} M \dot{w}]/2$, where $w = v - \gamma u$, agree throughout this family for matching clock initial data.

Proof. For an edge matrix $S > 0$, set $\kappa = S_{22}$ and $\gamma = -S_{12}/S_{22}$. Its normalized Schur complement gives $S_{11} = 1 + \kappa\gamma^2$. The change to u, w has determinant one and separates two positive quadratic forms. Completing the square gives the common minimum $-j^{\mathsf{T}} K^{-1} j/2$ and a nonnegative excess energy. Stationarity, Gaussian integration and the separated kinetic equations then give the stated identities. □

The same separation works for any common C^2 clock energy $H(u; j)$ with a finite Gibbs partition. Coercivity alone does not imply that partition condition. In particular, the resting-source energy $H = u^{\mathsf{T}} K u/2 + \sum_i j_i(e^{-u_i} - 1)$, $j_i \geq 0$, has positive Hessian, a unique minimum and a Gaussian lower bound ensuring integrability. Its stationary equation $Ku = je^{-u}$ therefore preserves the spatial ambiguity with the same proper-time coupling used for probe clocks.

Use one metric and one universal matter/Maxwell dictionary for every member:

$$ds^2 = -e^{-2u}c^2dt^2 + e^{2v}d\mathbf{x}^2, \qquad S_{\text{clock}} = -mc^2 \int d\tau, \qquad S_{\text{photon}} = S_{\text{Maxwell}}[g]. \tag{27}$$

An identical local gap ΔE has coordinate phase rate $e^{-u}\Delta E/\hbar$, with $\Delta E = c^2\Delta m$ for a rest-mass gap. Static exchange gives $\nu_r/\nu_e = e^{u_r - u_e}$, independent of γ. The optical index is $n = e^{u+v}$. For a calibrated exterior field $u = \mu/r$, $\mu = GM/c^2$, the infinity-to-infinity first-order deflection is $2(1+\gamma)\mu/b$ for fixed finite γ in the weak-field limit $|u|, |\gamma u| \ll 1$. The member $\gamma = -1$ is conformal and has exactly zero bending, while $\gamma = 1$ has the Einstein weak-field coefficient. The linearized exterior tensor is

$$G_{ij} = (\partial_i\partial_j - \delta_{ij}\Delta)(u - v).$$

For a nonzero compact source, $G_{ij} = 0$ selects $\gamma = 1$ in this class. All members have zero linearized exterior scalar curvature and a divergence-free linearized Einstein tensor. Scalar vacuum and Ward balance therefore do not supply this selector. The exponential $\gamma = 1$ metric is not asserted to solve the full nonlinear vacuum Einstein equation.

Sixteen independently replayed local histories cover both source laws, strengths, schedules and transverse seeds. Actual final writers feed clock and optical-index reads. Exact residuals and a maximum-principle barrier enclose the stationary fields, and the conformal and Einstein optical intervals are disjoint in every history. A compact C^1 density with a C^3 potential gives a controlled continuum sequence. For rays whose actual closest approach lies outside the source and $0 \leq p = (1+\gamma)\mu/r_{\min} \leq 10^{-3}$, direct differentiation of the exact optical integral gives $|\theta(p) - 2p| \leq 11p^2$. Numerical quadrature supplies a separate check, not a certified quadrature enclosure or a finite-observer radio error budget.

The retrospective measurement comparison retains the Pound–Snider static redshift result and all four overlapping Fomalont solar-radio reductions, including their measurement conventions and statistical errors. The bounded redshift systematic is kept separate. These compressed published results are not independent repetitions or a likelihood fit of raw observations. The same-input general-relativistic reference is evaluated without a fitted per-channel normalization. The construction identifies a missing physical tensor condition; it supplies no selected constitutive law, source-to-stress identification, physical calibration or autonomous apparatus history.

B.4 Finite energy and entropy on one interacting action

This field attachment uses the same supplied Weyl/contact CAR Hamiltonian as the bound-pair calculation, $H = d\Gamma[-(1/8)\sigma \cdot \sin k] + \sum_x n_{x0} n_{x1}$. On the open cube of 27 sites, its complete two-fermion space has dimension 1,431. Assign half of each Hermitian hopping bond to either endpoint and the contact energy to its site. Then

$$h_x = U D_x + \tfrac{1}{2} \sum_{b \ni x} H_b, \qquad \sum_x h_x = H, \qquad J_{x \to y} = i[h_x, h_y], \qquad \frac{dh_x}{dt} + \sum_y J_{x \to y} = 0.$$

Here t is Hamiltonian model time with the supplied convention $\hbar = 1$. Hermiticity and antisymmetry of the currents follow directly. Summing the identity over a region cancels all internal currents and gives its complete outward flux. Overlapping bond stars produce range-two currents; omitting them fails conservation. Exact Gaussian-integer matrices verify all 351 site pairs and 27 local equations, and deleting U changes 54 currents. This is interaction energy accounting, not a number current renamed as stress. The bond-energy division is stated explicitly and is not unique.

On this full finite algebra, choose the faithful density $\tau = e^{-\beta H}/Z$ with declared $\beta = \log(25/8)$. For every trace-zero Hermitian variation δ,

$$DS_\tau(\delta) = \beta \operatorname{Tr}(\delta H), \qquad S(\rho) - S(\tau) = \beta \operatorname{Tr}((\rho - \tau)H) - D(\rho \| \tau).$$

These follow from $-\log \tau = \beta H + \log Z$; no commutation of δ with τ is needed. The retained test uses a localized contact-pair perturbation that does not commute with the reference. This is a concrete noncommutative attachment; a finite-simplex entropy theorem has an incompatible domain and is not used. The full Gibbs reference is supplied, not prepared by native repair.

Signed CAR restriction to two adjacent sites keeps all local sectors $N = 0, 1, 2$ and gives an eleven-dimensional positive support. It does not make the state faithful on the full sixteen-dimensional local Fock space. For the restricted density, $K_A = -\log \tau_A$ has nonzero commutator with the bare internal H_A and fails even the best fit by $aI + bH_A + \mu N_A + \nu N_A^2$. Thus the global Gibbs identity is not silently transferred to a bare regional clock. This is a finite numerical countercontrol for the declared region and reference, not a universal regional modular obstruction.

An exact test strengthens the boundary for one specified regional algebra. On the fixed-N direct sum $\bigoplus_n (\mathcal{H}_A^n \otimes \mathcal{H}_{A^c}^{N-n})$, a faithful density admits a preserving unital completely positive retraction onto $\bigoplus_n (\mathcal{B}(\mathcal{H}_A^n) \otimes I)$ exactly when it has blocks $r_n \otimes \tau_n$, with $\operatorname{Tr} \tau_n = 1$. Multiplicative-domain identities remove cross-sector blocks and force this conditional product form. For a nonzero Gibbs parameter this requires block-additive H and hence $[H, N_A] = 0$. The actual adjacent-site region has 36 ordered crossing one-body entries of modulus $1/16$ in 54 modes, so $\|[N_A, H]\|_{\mathrm{HS}}^2 = (54 - 2)36/256 = 117/16$. No dense Gibbs logarithm is needed to exclude that exact retraction or its exact generator intertwining. This does not exclude partial traces, reduced modular clocks, approximate maps or different accessible domains. Where nested algebras do admit expectations preserving one faithful reference, uniqueness supplies their tower law. Independently selected regional references do not supply it.

A weaker observation interface does compose without factorization. For a CPTP state restriction T_{BS} and one faithful reference with $\rho_B = T_{BS}(\rho_S)$, define

$$E_{BS}(X) = \rho_B^{-1/2} T_{BS}(\rho_S^{1/2} X \rho_S^{1/2}) \rho_B^{-1/2}.$$

Its explicit Kraus matrices make it unital and completely positive, and $\operatorname{Tr}(\rho_B E_{BS} X) = \operatorname{Tr}(\rho_S X)$. For nested restrictions and their same reference reductions, cancellation of the intermediate square roots gives $E_{AB} E_{BS} = E_{AS}$. This finite Petz-transpose construction preserves complete POVMs

and arbitrary references. It generally neither fixes the embedded local algebra nor intertwines modular flows; its reference-dependent effects are not thereby implemented native local detectors. It supplies a compatible observation tower while leaving those stronger clock and access requirements explicit.

These constructions supply exact finite conservation and a quantum entropy map. They do not inhabit the Einstein premise bundle. Needed are a common symmetric Lorentz stress tensor, its geometric Ward identities and null charges, modular diamonds and their edge split, fixed-proper-volume area variation, Bianchi/geometry data and a controlled common refinement with physical normalization. In particular, finite Gibbs stationarity is not generalized-entropy stationarity at fixed proper volume. The complete source records and proof boundaries are in the repository [8]; these finite attachments close only the obligations on those declared algebras.

B.5 Entropy stationarity from a continuing native record

The native population interface admits an entropy calculation after record-adaptive repairs, without assuming an unread stationary continuation. Fix the binary total-two carrier and a faithful reference with weight w_q per configuration of initial core charge $q = n_0 + n_1$. The core pair observable is $c = n_0 n_1$. Let b be a complete native tape with bounded stopping length T_b, and let d_b count its compatible endpoints. Native branch inversion identifies each endpoint with an initial state. Assume the tape resolves q, and that any retained flag has probability $\beta > 0$ independent of the initial state.

Proposition B.6 (Counted native-record entropy envelope)**.** *For an initial population p, let p_B be its tape law conditional on that flag. Define*

$$F(p) = H(p_B) + \mathbb{E}_p \log d_B - \mathbb{E}_p T_B \log 2 - \log \beta.$$

Writing σ for the reference population, one has

$$F(p) - S(\sigma) = h_1(\langle q\rangle_p - \langle q\rangle_\sigma) + h_2(\langle c\rangle_p - \langle c\rangle_\sigma) - D(p_B\|\sigma_B),$$

where $h_1 = \log(w_0/w_1)$ and $h_2 = \log(w_1^2/(w_0 w_2))$. Hence the fixed-moment first variation vanishes at σ, and its Hessian is the negative semidefinite Fisher form of the tape response.

Proof. Injectivity and the retained fair coins give $\sigma_B(b) = d_b w_{q(b)} 2^{-T_b}/\beta$. Substituting this law in $D(p_B\|\sigma_B)$ leaves $-\sum_b p_B(b) \log w_{q(b)}$. State-independent flag probability makes its initial-state marginal p. Finally $-\log w_q = -\log w_0 + h_1 q + h_2 c$. The second variation of relative entropy is the Fisher form; directions invisible to this record map can remain flat. □

The charge-resolution hypothesis has a native readout. For E, the contrasts on seams $(0,1)$ and $(0,5)$ determine the three binary occupations at the start of that census: a common all-one offset is impossible at total load two. Retained prefix currents then recover the initial core, including after crossing repairs. No success conditioning is needed. For C, waiting for both coin values on each census seam gives a bounded service with state-independent success; all timeouts stay in the complete instrument. Unresolved timeout fibres have counted-entropy excess bounded by $(1-\beta)\log(w_{\max}/w_{\min})$. The expected coin cost must vary with p when stopping depends on records. Cap completions are marginals of the same conditional endpoint measure, so their shared predictions agree without independently resetting the source.

This is a population-record entropy theorem. It identifies neither the completion count nor the coin-cost subtraction with gravitational area, nor the core moments with physical volume or stress.

Its stationarity is a fixed-moment record stationarity, not the fixed-proper-volume generalized-entropy variation of (G4); joining the two requires an area, volume and stress attachment on the same process.

B.6 Regional first law from native return counts

The binary source supplies a finite regional entropy calculation from its own closed histories. Let C count the sixty seam/coin choices per input. The length-two reconstructed state is

$$\sigma = \frac{C^2}{171720} \geq mI, \qquad m = \frac{40}{4293}.$$

A two-port seam has fixed-load sectors with exterior dimensions $d_q = 45, 10, 1$. If R_q is the exterior partial trace in sector q, its regional density $R = \bigoplus_q R_q$ has trace one. The tracial completion $\mathcal{E}(\sigma) = \bigoplus_q R_q \otimes I_{d_q}/d_q$ gives

$$S_{\text{bulk}} = H(p) + \sum_q p_q S(R_q/p_q), \qquad S_{\text{edge}} = \sum_q p_q \log d_q, \qquad p_q = \operatorname{Tr} R_q.$$

The central Shannon term is in the bulk. The original full reference has nonzero cross-sector entries and differs from this completion. The edge multiplicities are not thereby gravitational area.

For $|f| \leq 1$ and $\operatorname{Tr}(C \operatorname{diag}(f) C) = 0$, weighting a return history at its midpoint by $1 + tf$ gives the faithful family $\rho_t = \sigma + tC \operatorname{diag}(f) C/171720$, $|t| < 1$. Its regional restriction is $R + tY$. On this same algebra, with $K = -\log R$,

$$\Delta S_{\text{bulk}} = t \operatorname{Tr}(YK) - D(R + tY \| R), \qquad 0 \leq D(R + tY \| R) \leq \frac{t^2 \|Y\|_{\text{HS}}^2}{2(1 - |t|)m}.$$

The bound follows by integrating the positive noncommuting entropy Hessian along the faithful segment. It controls the first law without a numerical differentiation assumption. Exact native counting supplies sixty-five independent real symmetric filter directions. One filter preserves all twelve mean occupations and has strictly positive bulk entropy derivative, while the bare seam cut count stays eight. No positive constant multiple of that static cut count cancels the derivative. These are postselected history ensembles with complete return and filter weights, and no causal quantum preparation is inferred. Fixed mean occupation is not fixed proper volume, so these identities supply neither responsive area nor fixed-proper-volume stationarity. Exact counts, rational logarithm bounds and independent tensor checks are in the repository [8].

A reference conditional on a recorded native current is another finite construction. The binary current instrument relaxes to $I_{66}/66$; a declared complete chart transducer, reduced conditional on the current alone, gives a faithful three-dimensional state. Conditioning additionally on the chart outcome gives a pure state, and two-record conditioning can reduce rank. Consequently the logarithm and first-law domain must follow the chosen reference support. The associated move-count clock has mean $5/33$ per attempt and asymptotic variance $1597/11979$, including correlations. Neither this count nor the conditional modular generator fixes a physical time, temperature, or gravitational area coefficient [3].

B.7 Regional entropy of a prepared interacting state

The open four-site Weyl/contact plaquette has eight signed canonical anticommutation modes, speed $c = 1/8$, contact coefficient $U = 1$ and inverse temperature $\beta = 1/32$. Its source preparation

executes 256 classical column records and retains every abort outcome. Write Σ for the unnormalized success field marginal, summed over those records, and $\sigma = \mathrm{Tr}_{\mathrm{out}}\,\Sigma$ for its signed restriction to the complete sixteen-dimensional matrix algebra on modes $(0,1,2,3)$, so that $\mathrm{Tr}\,\Sigma = \mathrm{Tr}\,\sigma = p \le 1$. Exact rational certificates bound the complete preparation error below 1.7×10^{-12} and give

$$\sigma \ge mI, \quad \Sigma \ge m_{\mathrm{full}} I, \quad m > \frac{1}{20}, \quad 16\, m_{\mathrm{full}} > \frac{1}{20}, \quad \|K - P_8 - aI\| \le E < 2 \times 10^{-9}, \quad K = -\log\sigma,$$

where P_8 is the eighth-order regional logarithm polynomial of the same action and a is the Gibbs normalization scalar. The polynomial retains the noncommuting boundary corrections of the reduced logarithm; it differs from β times the isolated two-site Hamiltonian.

Proposition B.7 (Two-sided regional entropy response)**.** *Let $X = X^*$ on the region, with $\mathrm{Tr}\,X = 0$ and $r = \|X\| \le 1/40$, and let V be the signed Fock factorization of the field into the region and its sixteen-dimensional complement. For $|t| \le 1$,*

$$\sigma + tX \ge (m - r)I > \frac{I}{40}, \qquad \Sigma_t = \Sigma + tV^*\Big(X \otimes \frac{I_{16}}{16}\Big)V \ge \Big(m_{\mathrm{full}} - \frac{r}{16}\Big)I > \frac{I}{640},$$

and Σ_t has signed regional restriction $\sigma + tX$ and trace p. With $S(A) = -\mathrm{Tr}(A \log A)$ and $D(\sigma + X\|\sigma) = \mathrm{Tr}((\sigma + X)(\log(\sigma + X) - \log\sigma))$,

$$S(\sigma + X) - S(\sigma) = \mathrm{Tr}(XK) - D(\sigma + X\|\sigma),$$

$$\tfrac{1}{2}\|X\|_{\mathrm{HS}}^2 \le D(\sigma + X\|\sigma) \le \frac{\|X\|_{\mathrm{HS}}^2}{2(m - r)} \le 20\|X\|_{\mathrm{HS}}^2,$$

$$|S(\sigma + X) - S(\sigma) - \mathrm{Tr}(XP_8)| \le E\|X\|_1 + \frac{\|X\|_{\mathrm{HS}}^2}{2(m - r)}.$$

For Hermitian trace-zero Δ, $h > 0$ and $l = m - h\|\Delta\| > 0$,

$$\left|\frac{S(\sigma + h\Delta) - S(\sigma - h\Delta)}{2h} - \mathrm{Tr}(\Delta P_8)\right| \le E\|\Delta\|_1 + \frac{h^2\|\Delta\|\,\|\Delta\|_{\mathrm{HS}}^2}{6l^2}.$$

Proof. Equal traces give the entropy identity. Differentiating $\log A = \int_0^\infty [(1+s)^{-1}I - (A+sI)^{-1}]\,ds$ at $A > 0$ gives $D\log_A[X] = \int_0^\infty (A + sI)^{-1}X(A + sI)^{-1}\,ds$, and the entropy Hessian is minus $g_A(X, X) = \mathrm{Tr}(X\, D\log_A[X])$. In an eigenbasis of A its coefficient on $|X_{ij}|^2$ lies between the reciprocals of the largest and least eigenvalues. Along $\sigma + tX$ the trace $p \le 1$ bounds the largest eigenvalue by one and the least eigenvalue is at least $m - r$. Integrating the Hessian twice gives the relative-entropy bounds. Trace duality gives the polynomial estimate, and a drops out because $\mathrm{Tr}\,X = 0$. For $f(t) = S(\sigma + t\Delta)$ and $R = (\sigma + t\Delta + sI)^{-1}$, one further differentiation gives $f'''(t) = 2\int_0^\infty \mathrm{Tr}(\Delta R\Delta R\Delta R)\,ds$. Schatten inequalities bound it by $\|\Delta\|\,\|\Delta\|_{\mathrm{HS}}^2/l^2$ on $|t| \le h$, and Taylor's integral remainder gives the symmetric estimate. □

The 255 matrices $E_{ii} - E_{15,15}$, $E_{ij} + E_{ji}$ and $i(E_{ij} - E_{ji})$ each have operator norm one, trace norm two and squared Hilbert–Schmidt norm two, and their Gram determinant is $16 \cdot 2^{240}$. They span every trace-zero Hermitian regional variation, including number-changing and noncommuting ones, and every bound holds on the parity-preserving subspace of dimension 127 and the number-preserving subspace of dimension 69. For one of these directions the symmetric estimator has error below $4 \times 10^{-9} + (1600/3)h^2$ on $0 < h \le 1/40$. The abort block is held fixed and contributes zero to entropy differences. The variations are field-marginal lifts at fixed success probability, without a record-resolved lift or an executed native intervention, and the plaquette is a source site region rather than a geometric diamond, so these identities supply neither responsive area nor fixed-proper-volume stationarity.

B.8 A complete native action of the regional modular polynomial

The same prepared region admits an executed modular clock. In the convention $\alpha_s(A) = \sigma^{is} A \sigma^{-is}$ the state unitary is $W_s = e^{isK}$. The polynomial has an exact real Pauli decomposition $P_8 = \sum_j b_j P_j$ with 39 nonzero terms, its scalar identity term among them, and $B = \sum_j |b_j| < 0.051$. It commutes with regional number and parity. For $h = s/n$, let $S(h)$ execute each factor $e^{ihb_j P_j/2}$ in listed order and then in reverse order, every factor being the source compiler's literal word of basis changes, parity controls, one phase rotation and uncomputation.

Proposition B.8 (Native regional modular actuator)**.** *For real s and integer $n \geq 1$,*

$$\|e^{isP_8} - e^{-isa} W_s\| \leq |s| E, \qquad \|S(s/n)^n - e^{isP_8}\| \leq \frac{B^3 |s|^3}{3n^2}.$$

At $s = 1/2$ and $n = 32$ the native word has 37,568 *pulses. It defines a complete map with successful output W and first-failure outputs F_f, $W^*W + \sum_f F_f^* F_f = I$. Up to the scalar phase, which cancels from every channel, its half-diamond distance from conjugation by $W_{1/2}$ is below 6.4×10^{-9} on all inputs, including the prepared columns, the four exterior field bits and an arbitrary untouched reference.*

Proof. Anti-Hermitian Duhamel and $\|K - P_8 - aI\| \leq E$ give the first bound. The palindrome obeys $S(-h) = S(h)^{-1}$ and $S'(0) = iP_8$, hence $S''(0) = S'(0)^2$, so $S(h)$ and e^{ihP_8} agree through second order. All factors are unitary, and the multinomial rule bounds both third derivatives by B^3. Taylor's remainder gives $B^3|h|^3/3$ per step and unitary telescoping gives the second bound. For each pulse with emitted generator G_e and ideal proper-code generator T_e, $\|e^{G_e} - e^{T_e}\| \leq \|G_e - T_e\|$, with the outward enclosure taken over all six processor inputs. Isometric telescoping bounds the literal word, with every failure output retained, by the sum d_{lit} of those errors. The total $d_{\mathrm{lit}} + B^3|s|^3/(3n^2) + |s|E$ has exact rational value near 6.385×10^{-9}: 5.34×10^{-12} for the literal word, 5.388×10^{-9} for the product and 9.92×10^{-10} for the bridge to the prepared logarithm. □

All sixteen regional input columns are executed. Continuing the preparation is exact channel composition: each unnormalized success column becomes $v_j' = (I_{\mathrm{ext}} \otimes W) v_j$, $j = 0, \ldots, 255$, with the eight preparation record bits, every earlier abort output and all first-failure labels retained and no source reset in between. The target W_s preserves σ, so the same bound controls regional stationarity. On the full plaquette a regional unitary can change correlations and need not preserve Σ, so it does not intertwine the regional modular group with the global one, in accord with Proposition B.9. The load, export and source services are the admitted simulator specification and s is dimensionless, so this operational clock identifies no geometric flow, stress charge or physical duration. The certificates and replays are in the repository [8].

B.9 Finite modular covariance and compressed responses

Geometric modular covariance requires a comparison beyond the existence of a finite modular clock. The following finite identity states a necessary boundary of that comparison, independently of the repair scheduler.

Proposition B.9 (Finite modular dilation and its compression boundary)**.** *Let G, P be Hermitian matrices on a finite Hilbert space, let $U(s) = e^{isG}$, and let $\kappa \in \mathbb{R}$. Then*

$$\|i[G,P] - \kappa P\|_{\mathrm{HS}}^2 = \|[G,P]\|_{\mathrm{HS}}^2 + \kappa^2 \|P\|_{\mathrm{HS}}^2, \qquad \|U(s) P U(s)^* - e^{\kappa s} P\| \geq |e^{\kappa s} - 1| \, \|P\|.$$

For an orthogonal projection Q, write $G_Q = QGQ$, $P_Q = QPQ$ and $B_Q = i(QG(1-Q)PQ - QP(1-Q)GQ)$. Then

$$Q\,i[G,P]Q = i[G_Q,P_Q] + B_Q, \qquad \|Q(i[G,P]-\kappa P)Q\|_{\mathrm{HS}} + \|B_Q\|_{\mathrm{HS}} \geq |\kappa|\,\|P_Q\|_{\mathrm{HS}}.$$

Proof. Cyclicity gives $\mathrm{Tr}(P\,i[G,P]) = 0$, so the cross term in the squared Hilbert–Schmidt norm vanishes. Unitary invariance of the operator norm and the reverse triangle inequality give the finite-parameter bound. Inserting $Q + (1-Q)$ into each product gives the compressed identity. Apply the first identity to the compressed matrices, then the triangle inequality, for the last bound. □

For the native faithful reference with per-configuration core weights $(2/225, 1/36, 2/45)$, take $G = \log\sigma$; thus s is the dimensionless modular parameter. Choose configurations a, b with core occupation zero and two, respectively. On their two-state corner declare

$$P = \begin{pmatrix} 1 & i/2 \\ -i/2 & 1 \end{pmatrix}, \qquad Q = |\psi\rangle\langle\psi|, \qquad |\psi\rangle = (|a\rangle + |b\rangle)/\sqrt{2},$$

and extend P by zero. It is positive, with nonzero eigenvalues $1/2, 3/2$. For $\kappa = \log(5)/2$, the projected first derivative is exactly κP_Q, supplied entirely by B_Q; the internal compressed commutator is zero. Nevertheless

$$\langle\psi|U(s)PU(s)^*|\psi\rangle = 1 + \tfrac{1}{2}\sin(s\log 5) \neq e^{s\log(5)/2}$$

in general. At $s = 1/\log 5$ the two values are approximately 1.420735 and 1.648721. The declared probe is not identified with physical null momentum.

Consequently a fixed finite Hilbert algebra cannot supply nonzero global modular dilation. A spectral projection of G has $B_Q = 0$ and retains the same obstruction. Restricted continuum comparisons must specify states, observable domains and finite parameter windows, retain the compression boundary, and control the actual responses on one refining family. Unbounded continuum generators are not excluded by these bounded-operator statements.

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